DATA4100

Data Visualisation and Communication

Lesson 2

Statistical methods for summarising and visualising data

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DATA4100 Roadmap

Week 1 Interactive, real time visualisation for business	Week 2 Statistical methods for summarising and visualising data	Week 3 Performing analytics, UX and report writing	Week 4 Using IT systems to assist data visualisation
Week 5 Communication of visualisations and Storytelling	Week 6 Making Visualisations more effective	Week 7 Revision Week	Week 8 In-Class Assessment
Week 9 Advanced Visualisations and comparing platforms	Week 10 Transforming data and drill through dashboards	Week 11 Python and comparing visualisation platforms	Week 12 Modelling and dashboards in python

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Lesson Learning Outcomes

1	Explore common statistical methods and how we visualise the results
2	Analyse data using Power BI for informed decision-making

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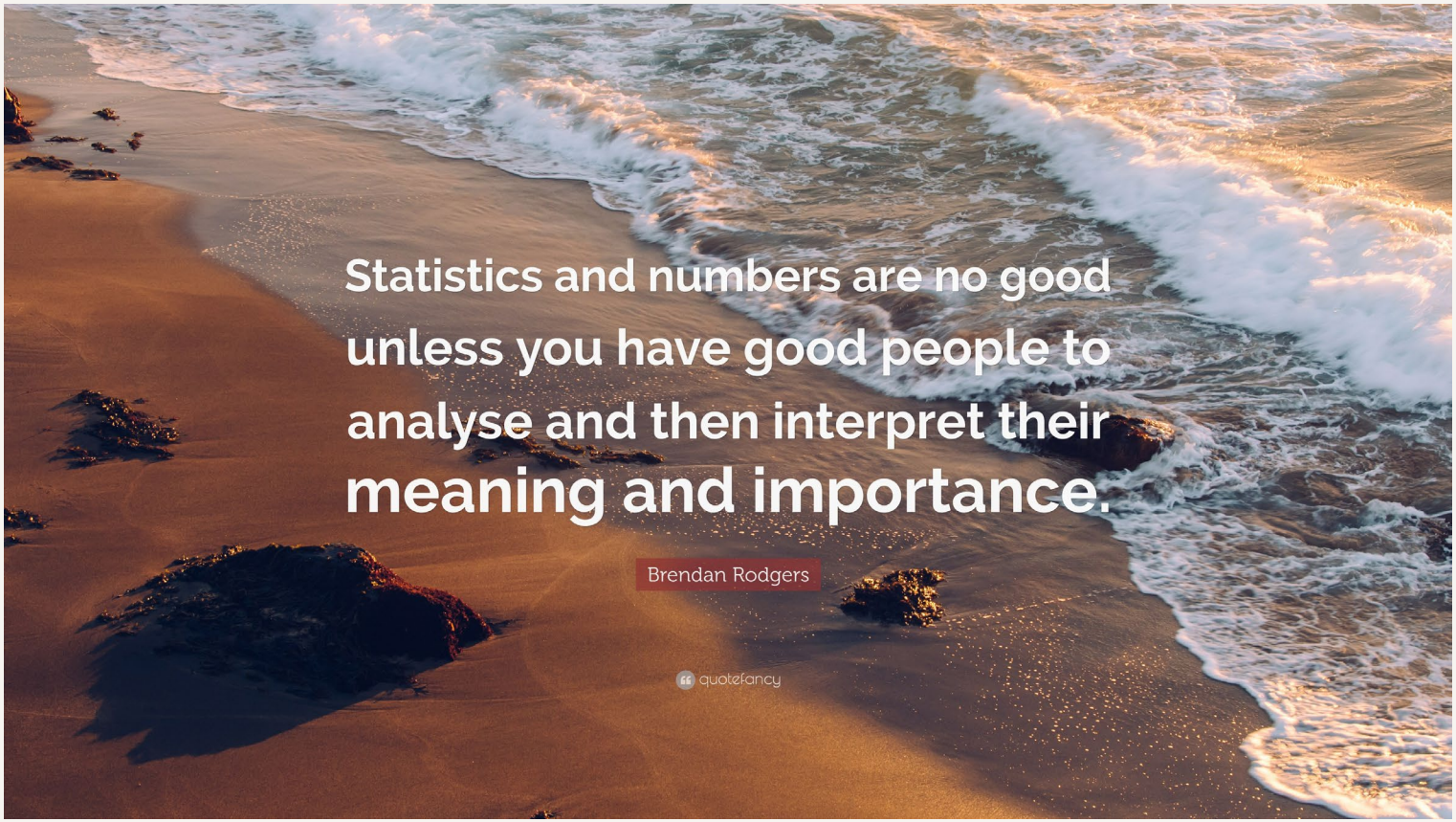
This Topic's Big Idea



Brendan Rodgers Quote: "Statistics and numbers are no good unless you have good people to analyse and then interpret their meaning and importanc..." (quotefancy.com)

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Download the sample dataset

- Download the dataset for “Superstore-Sales-Analysis” from Week 2 reading list.

- Go to ***GitHub*** (using the URL on MyKBS) > Choose ***Datasets*** > And then ***Superstore Dataset.csv***

- Tip: Click on the download icon to download the dataset as a csv file (see next slide for details).

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Business Case

- Superstore is a fictional retail company in the United States that sells furniture, office supplies, and technology products.

- As a Business Analyst, your role is to identify weaknesses and opportunities within the business to enhance growth and profitability.

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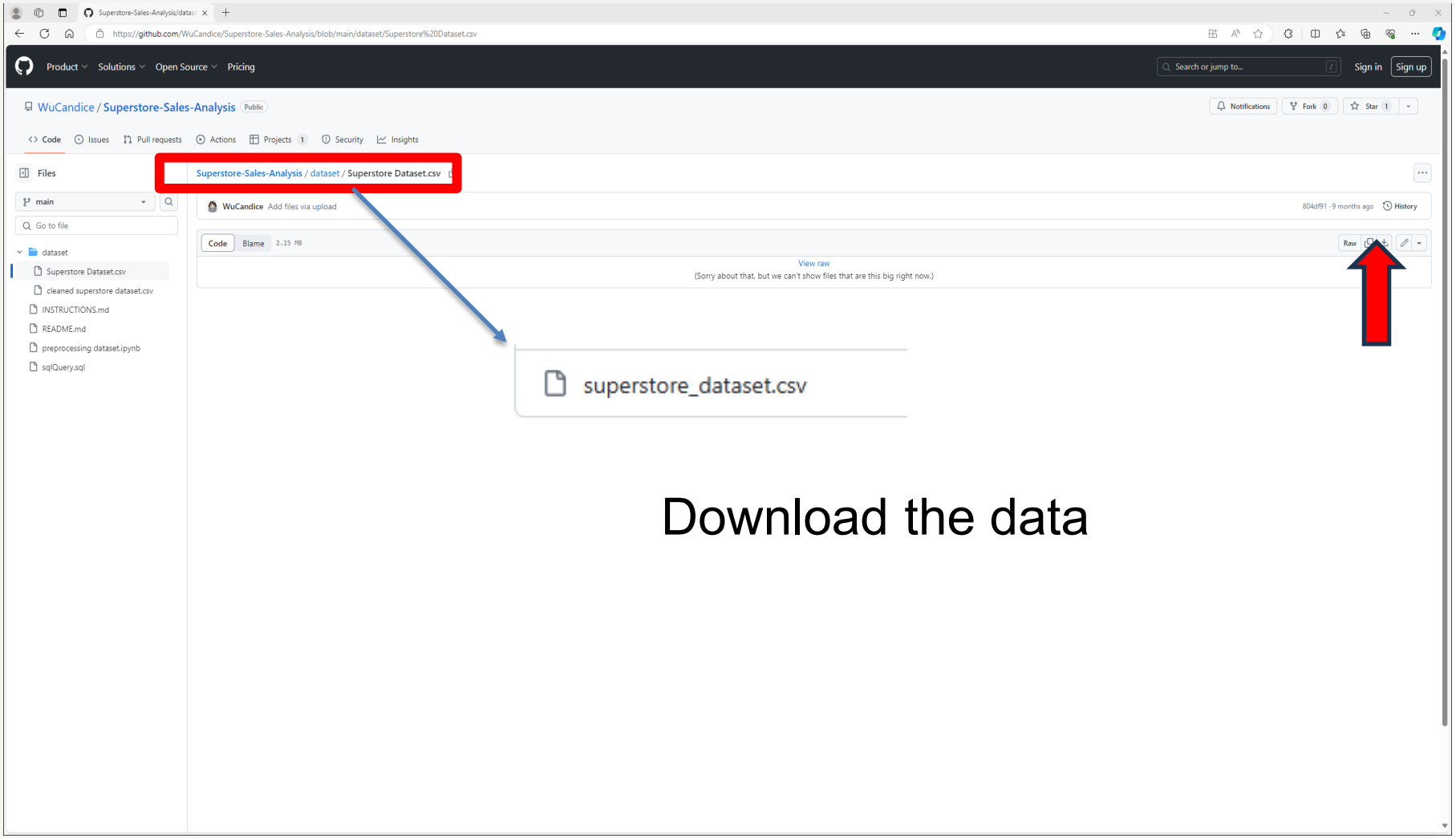
Activity: GitHub view



Download the data

Open the data in Excel for preliminary observations

Explore the dataset provided and consider what types of questions you could answer.

•Consider sales trends, customer demographics, product performance, and regional sales patterns.

•How can this data help Superstore improve its business strategy and operations?

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Example business questions

Using this dataset, we encourage you to explore the following questions to uncover valuable insights and strategic opportunities for Superstore:

1. How do sales and order volumes vary across the four regions (Central, East, South, West)?

2. Which subcategory performed best in terms of sales?

3. What are the top-performing products in the three main categories (Office Supplies, Furniture, Technology)?

4. Are any regions or cities showing growth potential in specific product categories?

5. How do customer demographics influence purchasing decisions and preferences?

- Form groups and create three business questions of your own.

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Activity: Superstore Dashboard

Now you've formulated your business questions, it's time to turn them into an interactive dashboard.

1. Design Your Dashboard:

- Create visuals to answer your business questions.
- Give your dashboard a clear, creative title reflecting the story you want to tell.

2. Think About Audience

- Who will use this dashboard? (e.g. sales managers, executives, operations teams, marketing).
- What decisions should they be able to make after viewing it?

3. Make It Engaging

- Use at least three different visual types (e.g. bar chart, map).
- Add interactivity (filters/slicers) so your audience can explore.

4. Reflect

Ask: *Does the dashboard clearly answer the questions? Would the intended audience find it useful?*

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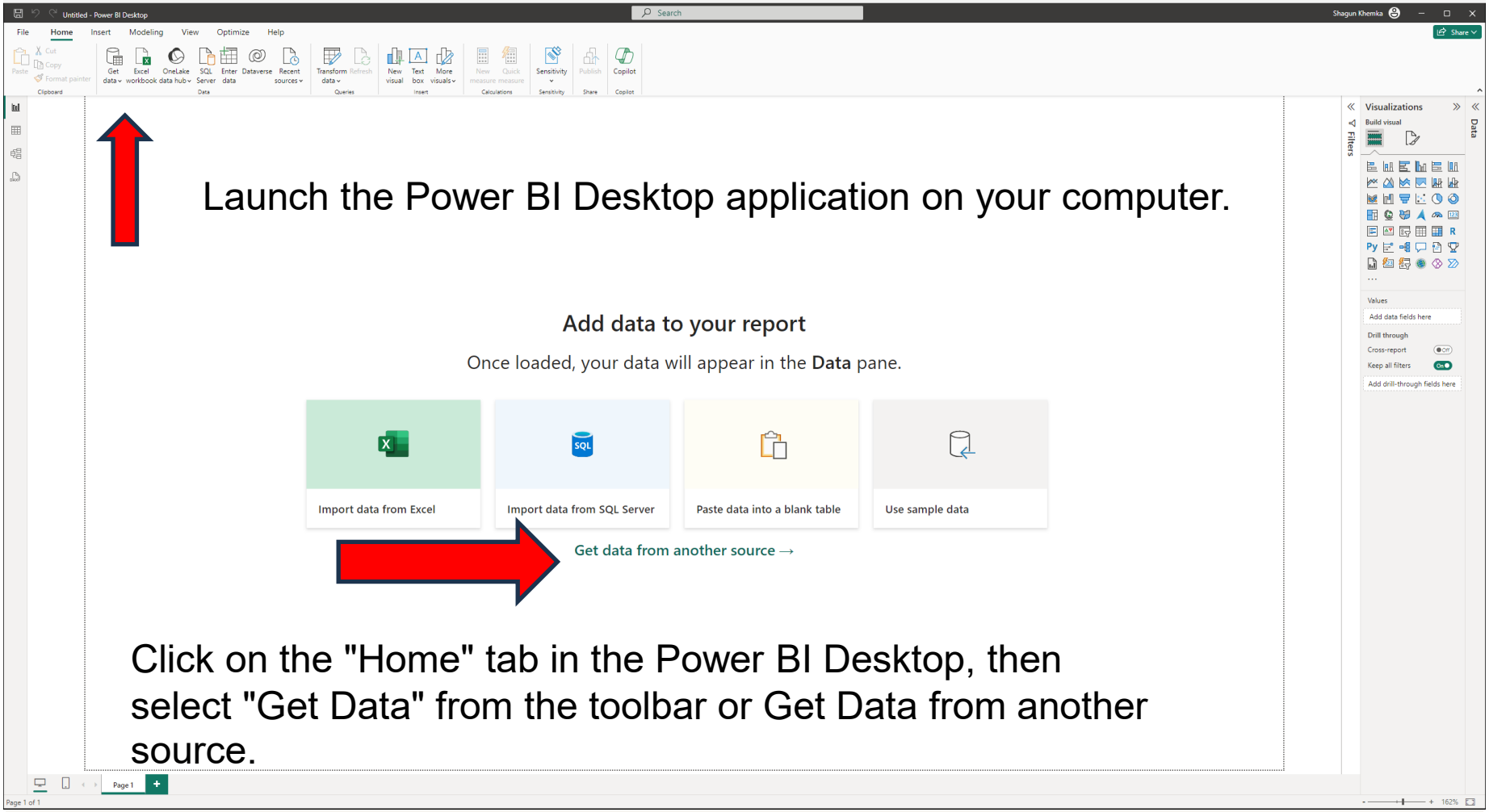
Open Power BI Desktop



Launch the Power BI Desktop application on your computer.

Click on the "Home" tab in the Power BI Desktop, then select "Get Data" from the toolbar or Get Data from another source.

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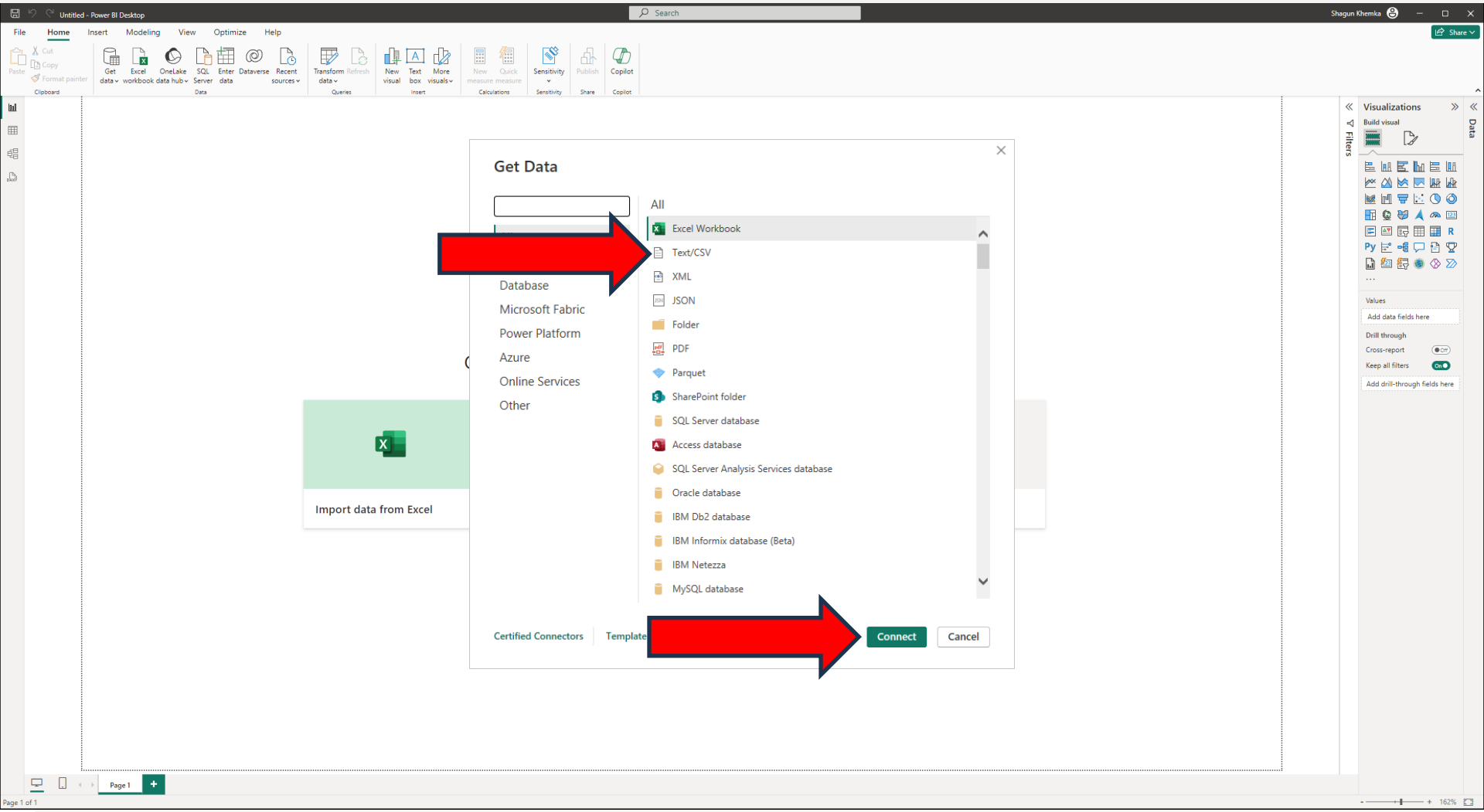
Choose your type of data set



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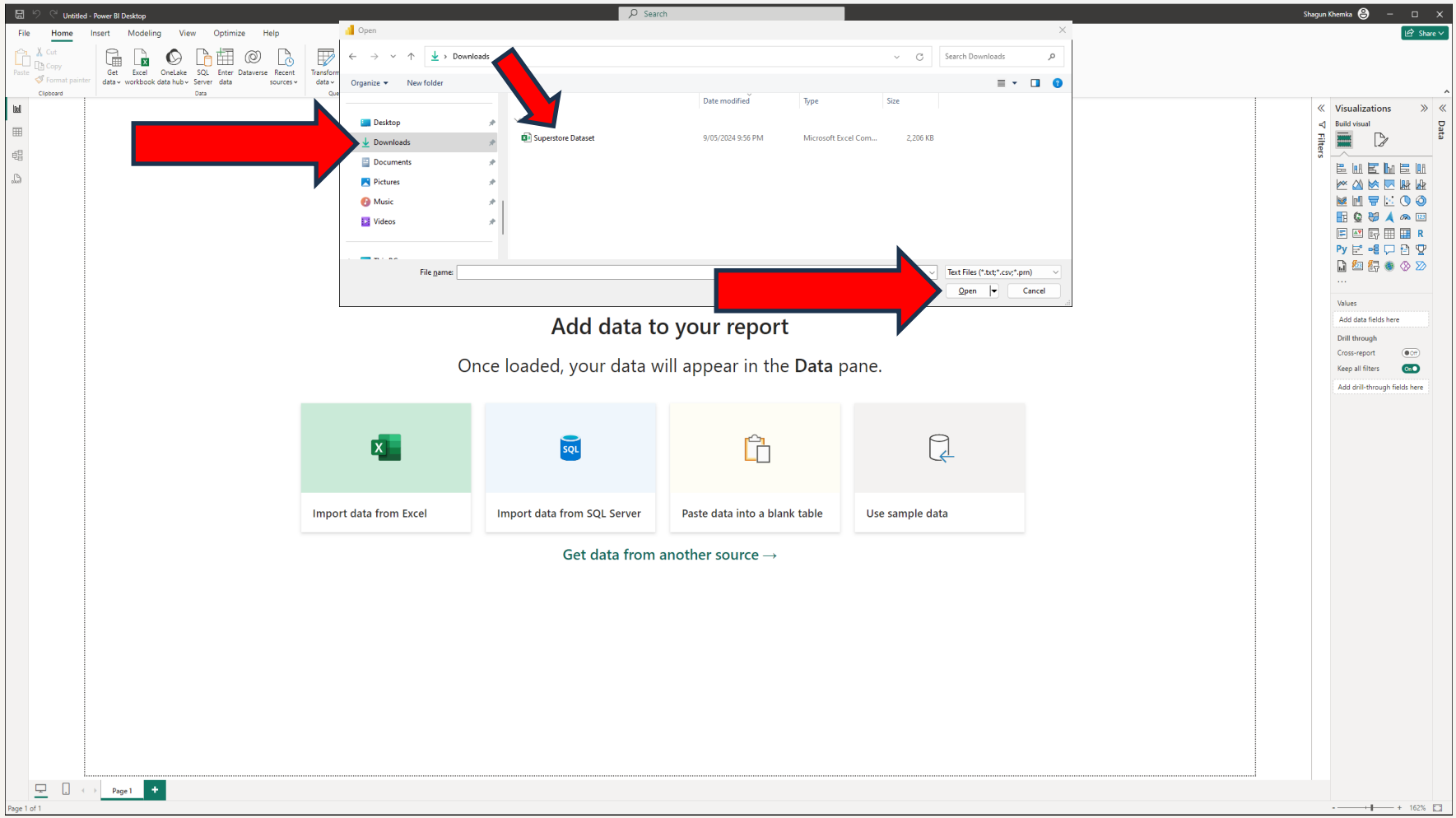
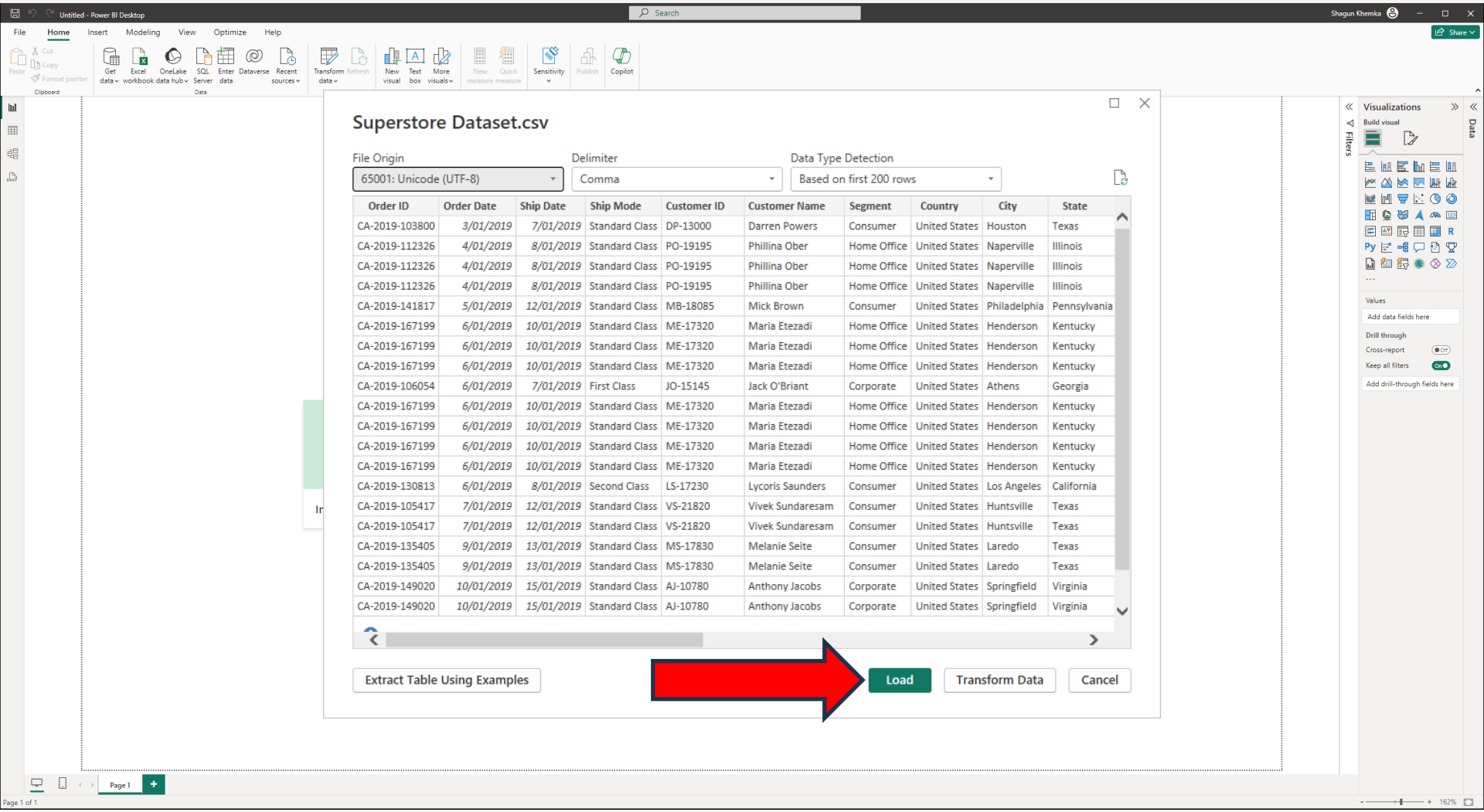
Access the data set



Load the data set



Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	State
CA-2019-103800	3/01/2019	7/01/2019	Standard Class	DP-13000	Darren Powers	Consumer	United States	Houston	Texas
CA-2019-112326	4/01/2019	8/01/2019	Standard Class	PO-19195	Phillina Ober	Home Office	United States	Naperville	Illinois
CA-2019-112326	4/01/2019	8/01/2019	Standard Class	PO-19195	Phillina Ober	Home Office	United States	Naperville	Illinois
CA-2019-112326	4/01/2019	8/01/2019	Standard Class	PO-19195	Phillina Ober	Home Office	United States	Naperville	Illinois
CA-2019-141817	5/01/2019	12/01/2019	Standard Class	MB-18085	Mick Brown	Consumer	United States	Philadelphia	Pennsylvania
CA-2019-167199	6/01/2019	10/01/2019	Standard Class	ME-17320	Maria Etezadi	Home Office	United States	Henderson	Kentucky
CA-2019-167199	6/01/2019	10/01/2019	Standard Class	ME-17320	Maria Etezadi	Home Office	United States	Henderson	Kentucky
CA-2019-167199	6/01/2019	10/01/2019	Standard Class	ME-17320	Maria Etezadi	Home Office	United States	Henderson	Kentucky
CA-2019-106054	6/01/2019	7/01/2019	First Class	JO-15145	Jack O'Briant	Corporate	United States	Athens	Georgia
CA-2019-167199	6/01/2019	10/01/2019	Standard Class	ME-17320	Maria Etezadi	Home Office	United States	Henderson	Kentucky
CA-2019-167199	6/01/2019	10/01/2019	Standard Class	ME-17320	Maria Etezadi	Home Office	United States	Henderson	Kentucky
CA-2019-167199	6/01/2019	10/01/2019	Standard Class	ME-17320	Maria Etezadi	Home Office	United States	Henderson	Kentucky
CA-2019-167199	6/01/2019	10/01/2019	Standard Class	ME-17320	Maria Etezadi	Home Office	United States	Henderson	Kentucky
CA-2019-130813	6/01/2019	8/01/2019	Second Class	LS-17230	Lycoris Saunders	Consumer	United States	Los Angeles	California
CA-2019-105417	7/01/2019	12/01/2019	Standard Class	VS-21820	Vivek Sundaresam	Consumer	United States	Huntsville	Texas
CA-2019-105417	7/01/2019	12/01/2019	Standard Class	VS-21820	Vivek Sundaresam	Consumer	United States	Huntsville	Texas
CA-2019-135405	9/01/2019	13/01/2019	Standard Class	MS-17830	Melanie Seite	Consumer	United States	Laredo	Texas
CA-2019-135405	9/01/2019	13/01/2019	Standard Class	MS-17830	Melanie Seite	Consumer	United States	Laredo	Texas
CA-2019-149020	10/01/2019	15/01/2019	Standard Class	AJ-10780	Anthony Jacobs	Corporate	United States	Springfield	Virginia
CA-2019-149020	10/01/2019	15/01/2019	Standard Class	AJ-10780	Anthony Jacobs	Corporate	United States	Springfield	Virginia

Data fields

The superstore dataset includes the following columns:

- **Order Details**: order_id, order_date, ship_date
- **Customer Information**: customer
- **Product Details**: manufactory, product_name
- **Segmentation**: segment, category, subcategory
- **Geographical Information**: region, zip, city, state, country
- **Financial Metrics**: discount, profit, quantity, sales, profit_margin

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Statistical methods

Before launching into analyses, let's

1. review statistical methods and associated charts/tables
2. identify possible fields that are suitable for each method

- Common methods and charts
  - Descriptive Statistics (tables, line charts, bar charts, pie charts and doughnuts, bubble charts)
  - Time series and forecasting (line charts over time/dates)
  - Correlation (tables/matrices and heatmaps)
  - Geographical methods (maps)
  - Regression (scatter and trend line of best fit)
  - Distributions (histograms, curves, density plots)

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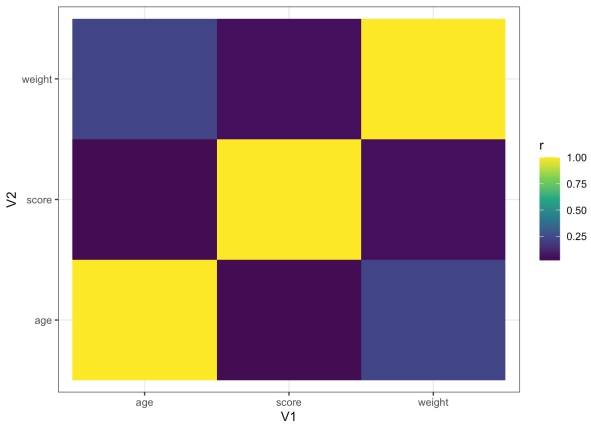
Possible combinations of variables for charts

- Descriptive Statistics (tables, line charts, bar charts, pie charts and doughnuts, bubble charts)
  - Financial metrics per segment, per larger region or per product
- Time series and forecasting (time series line charts)
  - Order date and some financial metrics
- Correlation (tables/matrices and heatmaps)
  - Financial metrics
- Geographical methods (maps)
  - City, state, country per segment or per product
- Distributions (histograms, curves, density plots)
  - Financial metrics



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Descriptive Statistics

Recall from DATA4000, Measures of location yield information about the centre of a data set.

Common Measures of Location

- **Mean** (average)

  $$\frac{\text{Sum of all values}}{\text{number of values}}$$

- **Median**

  Middle value (of ordered data)

- **Mode**

  (most frequently occurring value)

http://onlinestatbook.com/2/introduction/descriptive.html

Mean (Average) Example

Mean = Sum ($\sum$) all values in the data set and divide it by the number of values in the data set (n).

Example data: n = 16 a sample of arrival times in **minutes**
9 14 11 10 10 8 10 9 14 8 15 12 9 9 7 9

Sum all values then divide the sum by 16

Σx_i=9+14+11+10+10+8+10+9+14+8+15+12+9+9+7+9=164

Mean = 164/16 = 10.25 minutes **(10 minutes 15 seconds)**

In excel the command is =average(first cell:last cell)

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Mode and Median

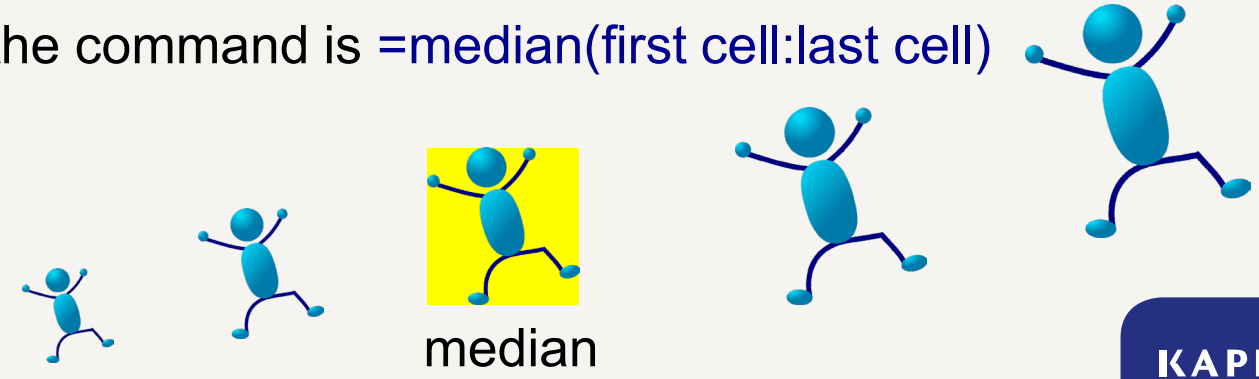
Definition: An **ordered array** of values is a list of values (data) arranged from the smallest to the largest, e.g. 1 1 1 2 5 7 7 9 12 15 26

- Mode is the **most frequently occurring value** in a data set, i.e. 1 is the mode (it occurs three times)

In excel the command is =mode(first cell:last cell)

- The median is the middle value (midpoint) in an ordered array of numbers, if n is an ODD number, otherwise it is the mean of the two middle values. In this case the median = 7.

In excel the command is =median(first cell:last cell)



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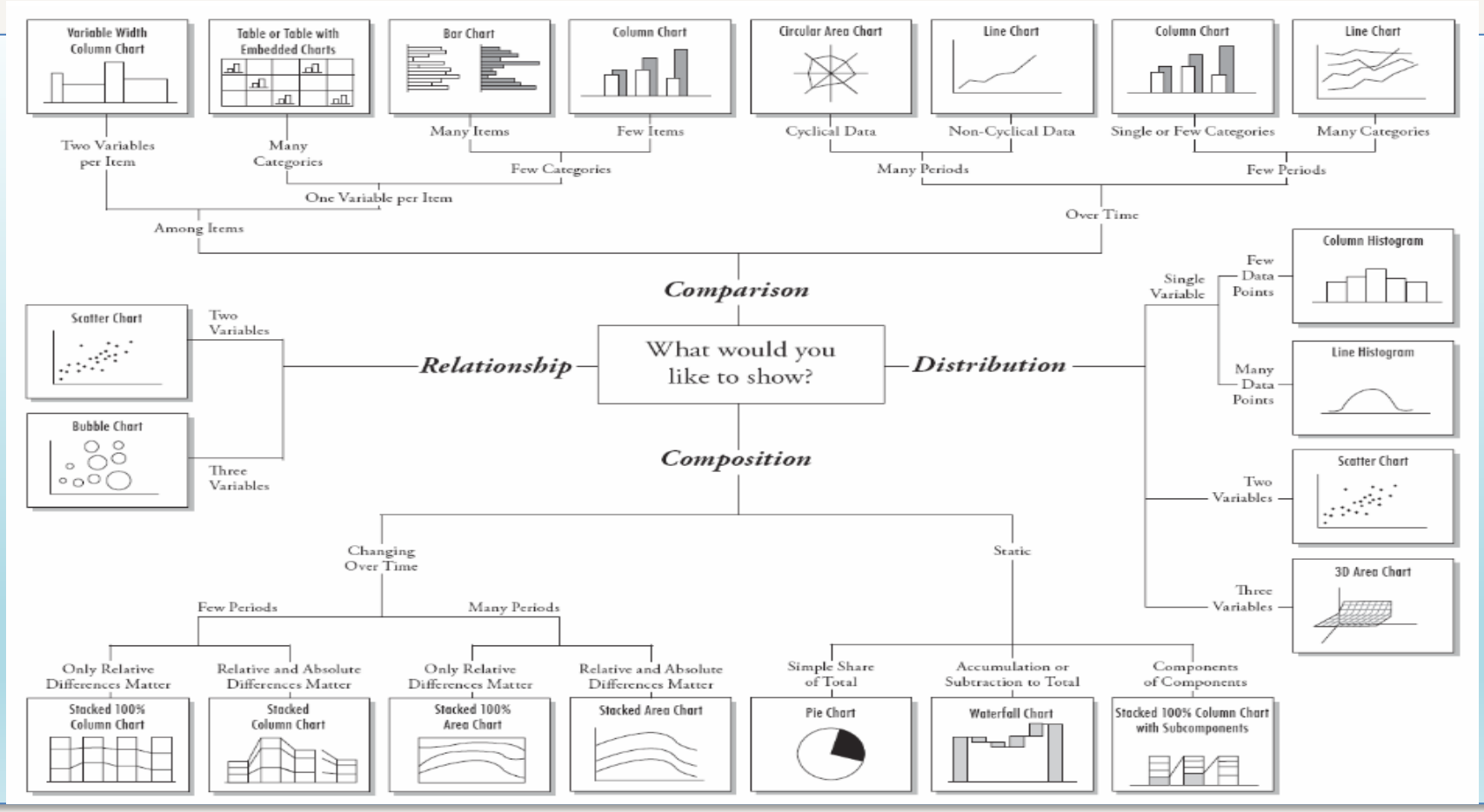
Which visualisation do I need?



Refer to above chart and guess which visualisation is needed to communicate the following:

1. The percentage of students of each nationality at your campus
2. The relationship between class attendance and grades
3. The distribution of final grades

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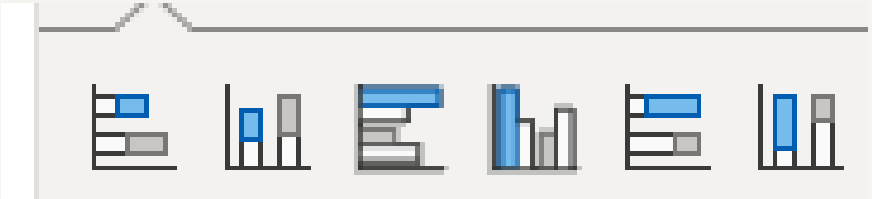
Bar and column charts
(Simple, Stacked, clustered)

Bar chart: One bar per category

- Height of bar is the frequency (counts) per category
- Width of bars should be equal
- Need gaps between rectangles to show that bars can be placed in different orders with categorical data.
- Satisfy the ‘area principle” of a chart: the area of a graph should equal the size of the data it is displaying.

- Icons in Power BI



- Stacked have more than one measure per bar/column
- Clustered are similar, however with the bars/columns are beside each other

- Run your pointer over the chart to see the names

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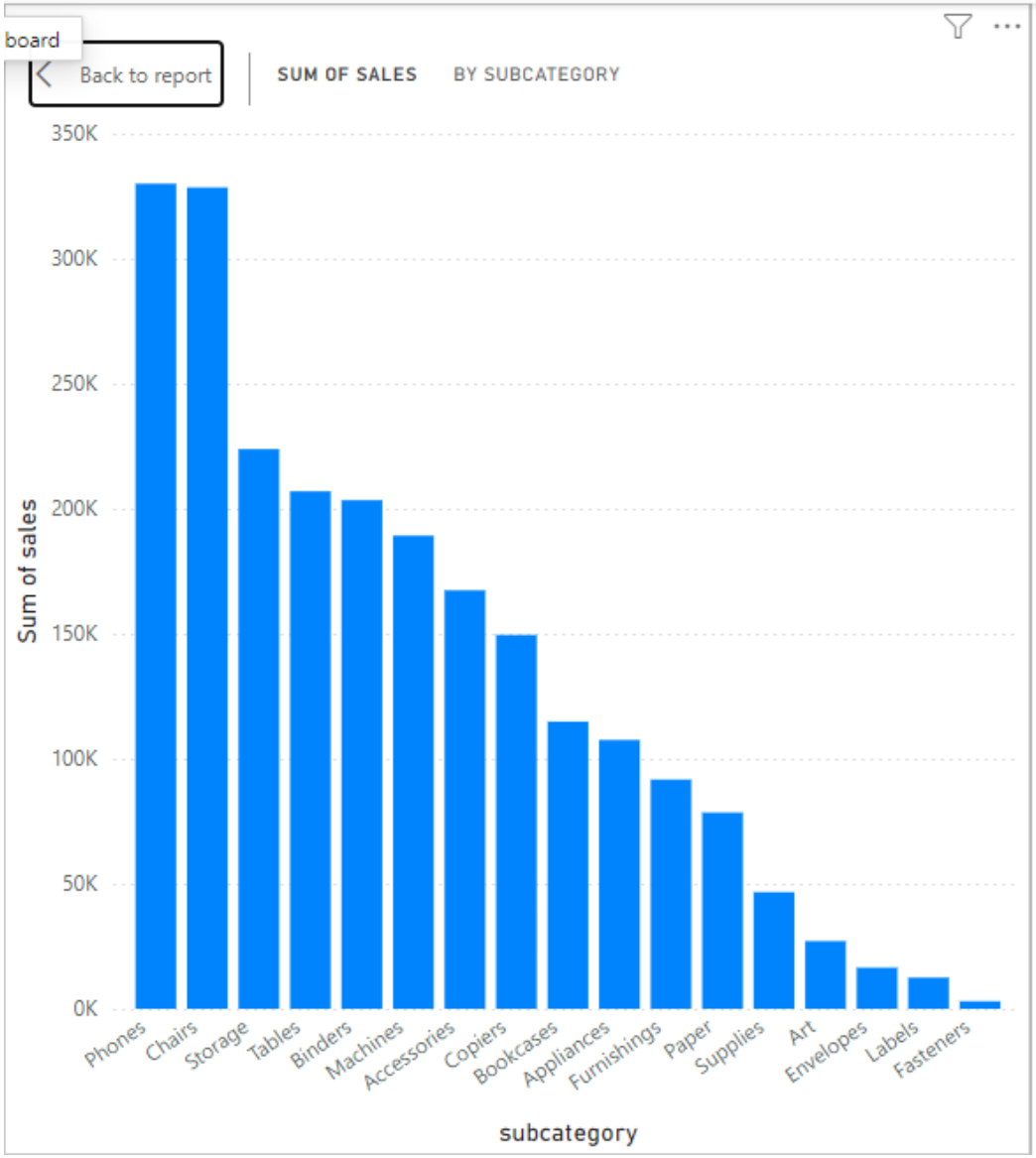
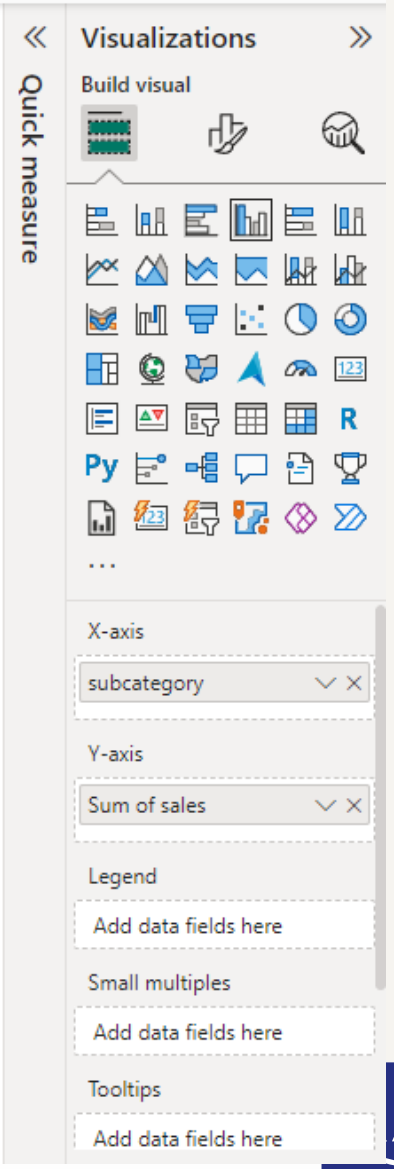
Simple Bar and column charts



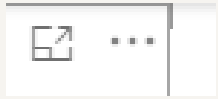


Select clustered columns chart

Drag Subcategory to X axis

Drag sum of sales to Y axis

Click on Focus mode (top RHS of chart frame)



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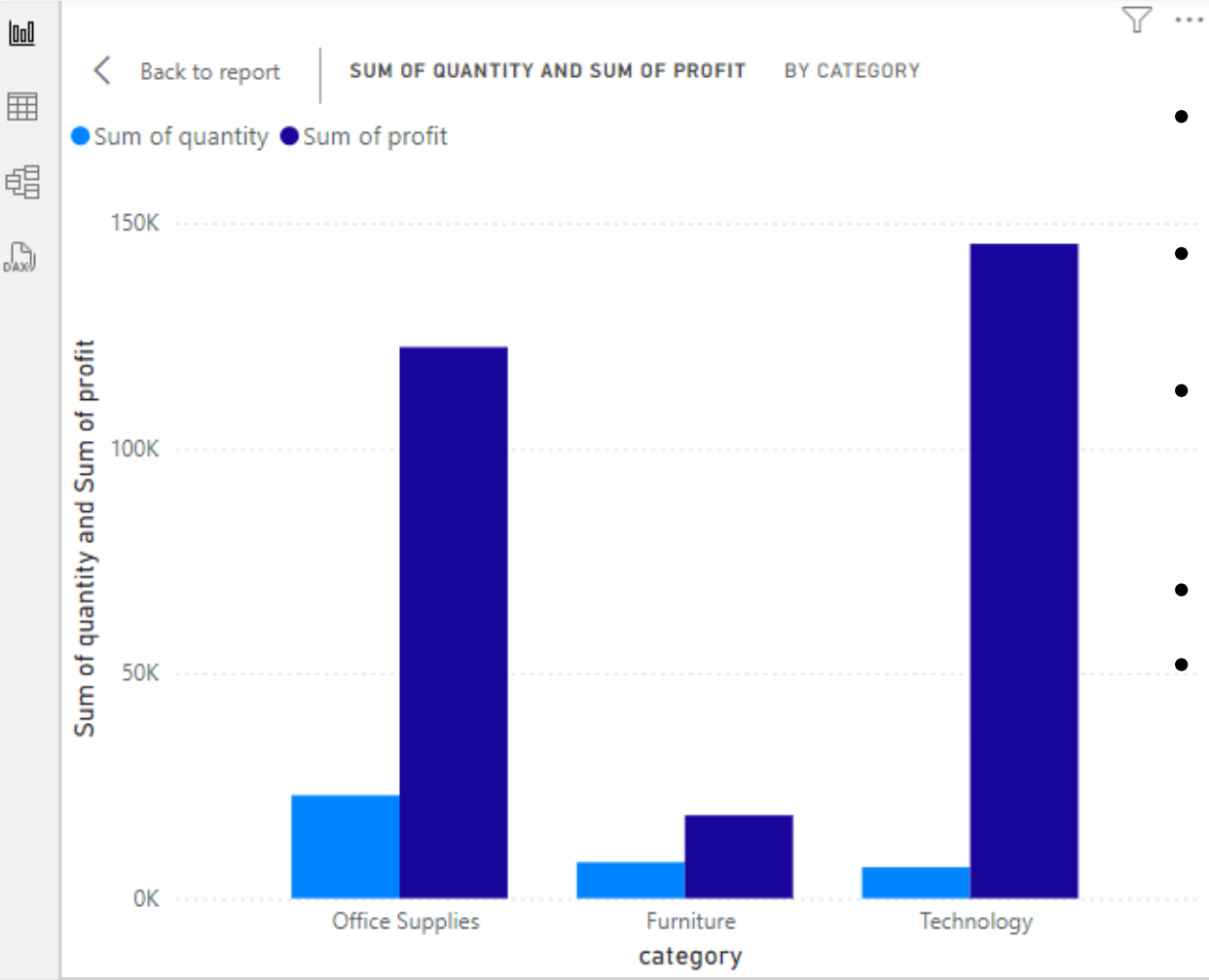
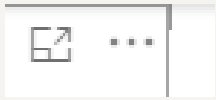
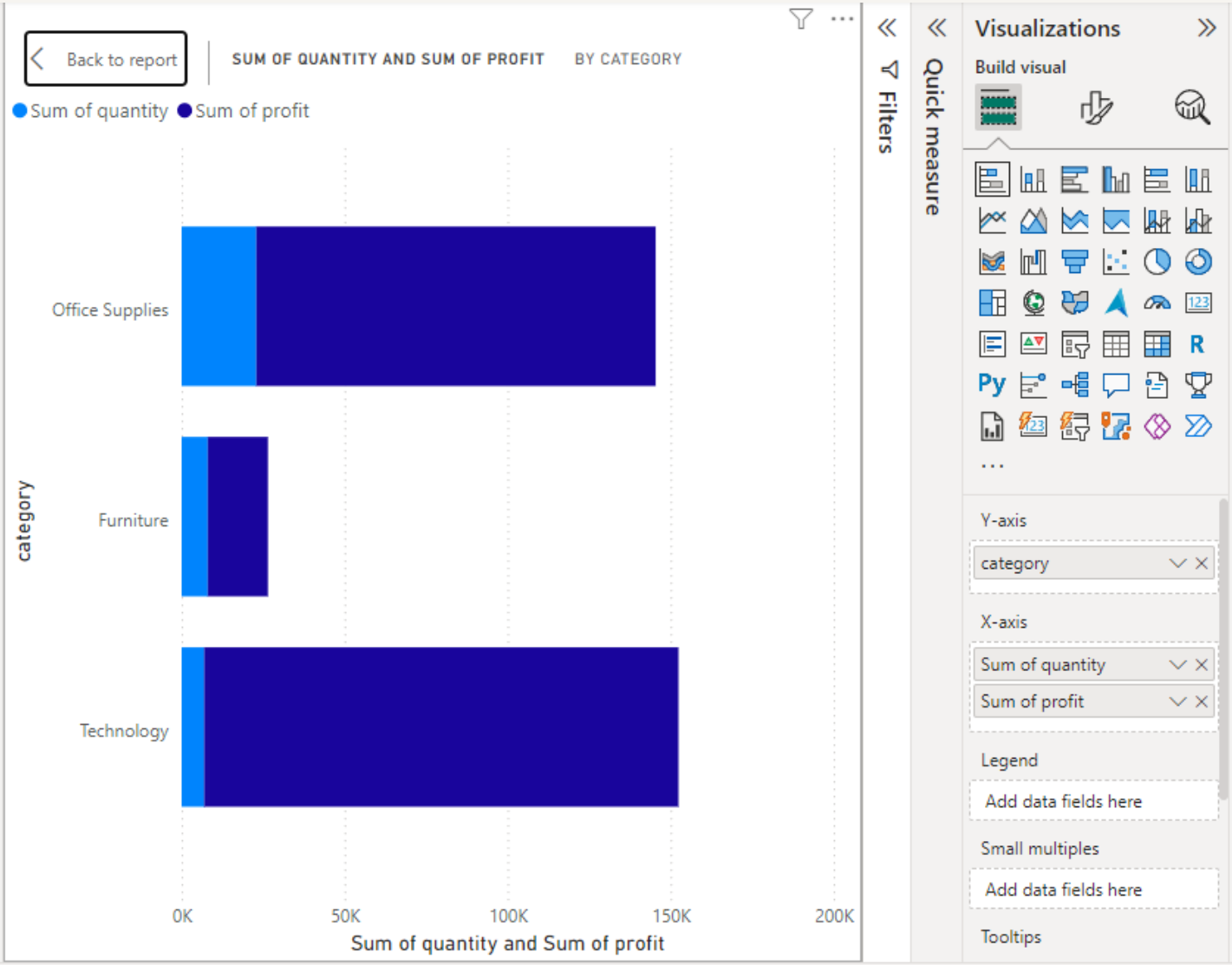
Bar and column charts



- Select clustered columns chart
- Drag Category to X axis
- Drag sum of Quantity and sum of profit to Y axis
- Click on Focus mode
- (top RHS of chart frame)

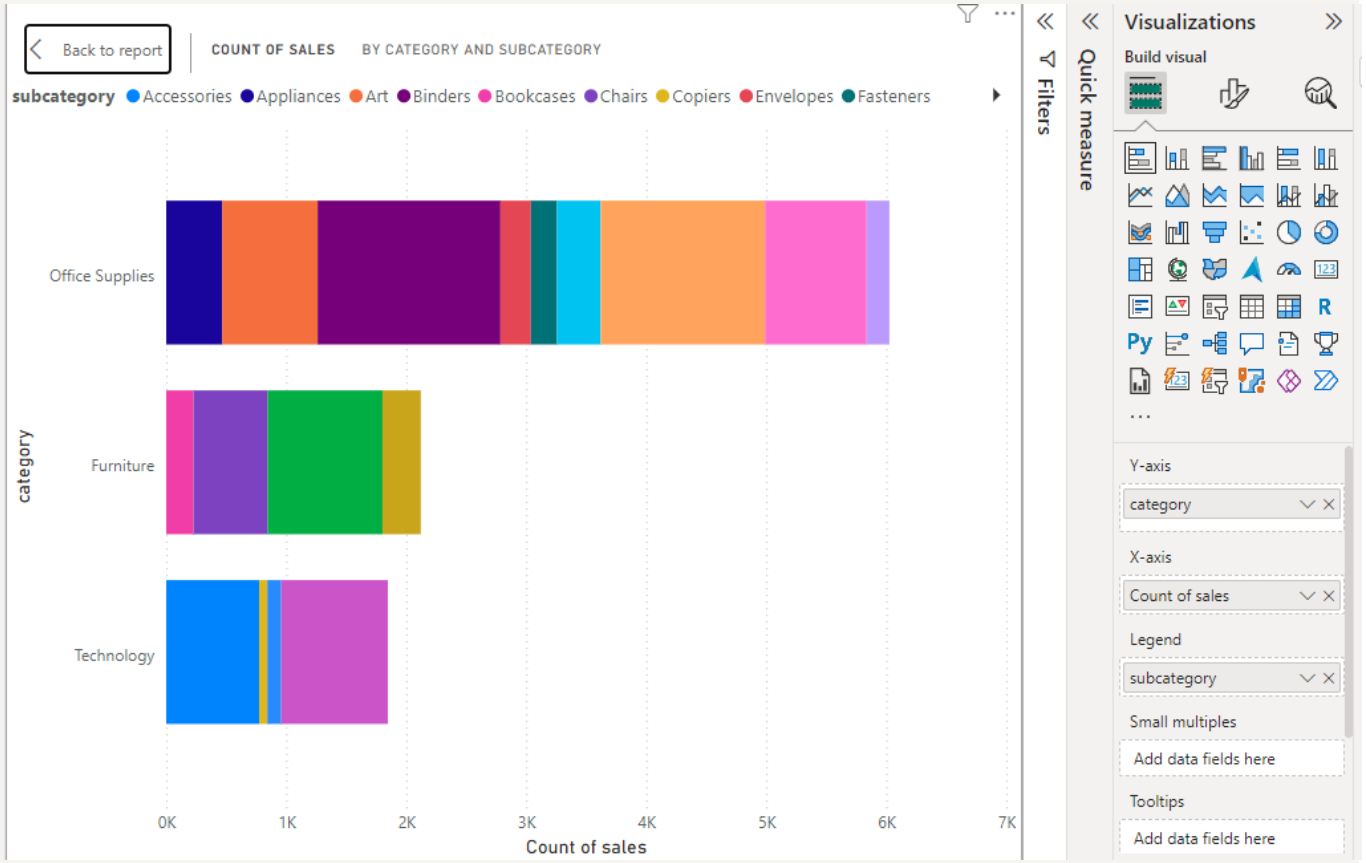


Change to stacked bar chart by clicking on the icon



Try a stacked bar chart with category and subcategory

- Select stacked bar chart
- Drag Category to Y axis
- Drag count of sales to X axis
- Drag subcategory to Legend
- Click on Focus mode



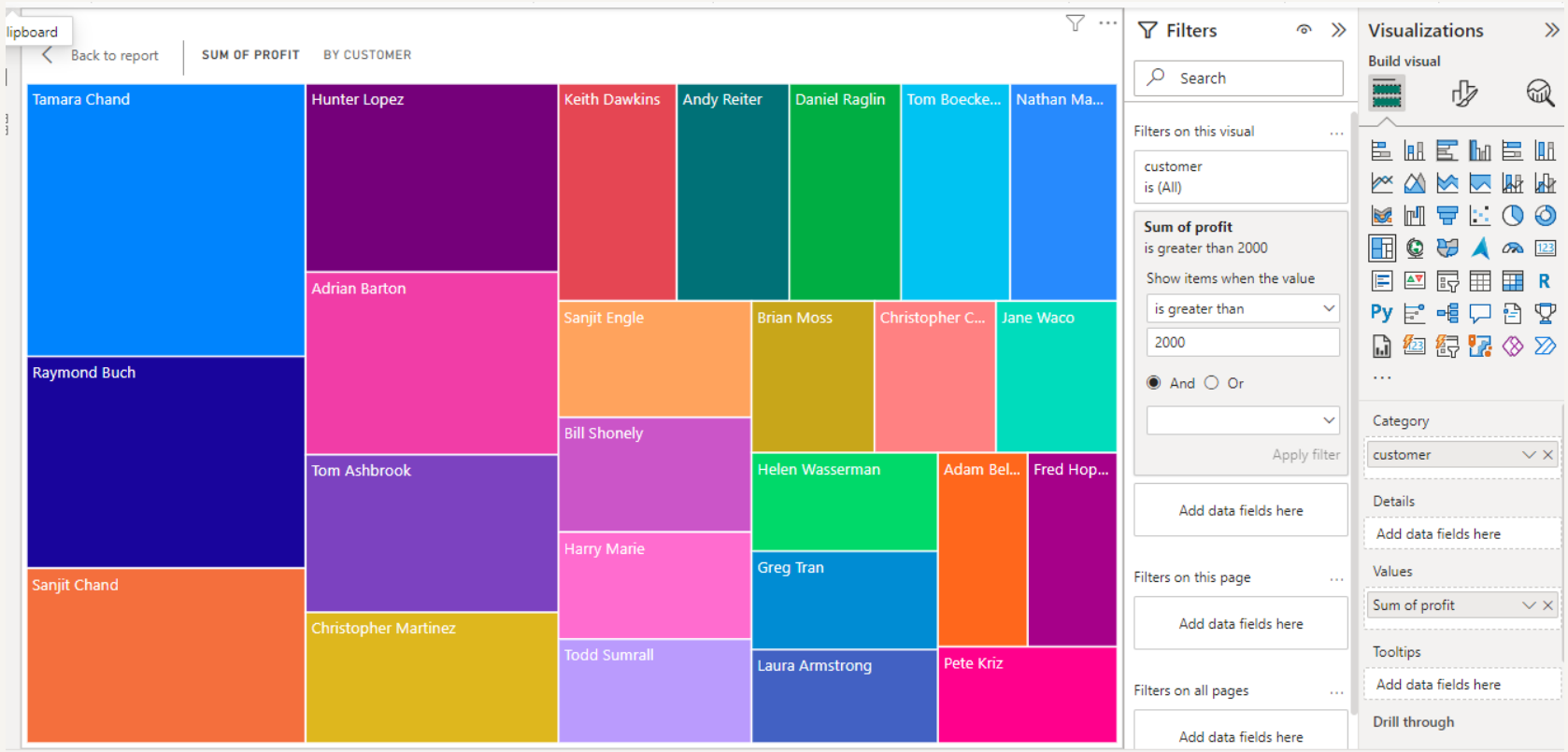
Treemap

- Select treemap
- A treemap shows variations in terms of squares, the best is the largest square
- Drag customer to category, and sum of profit to values

You can filter out the smaller values via the filters pane
See next slide

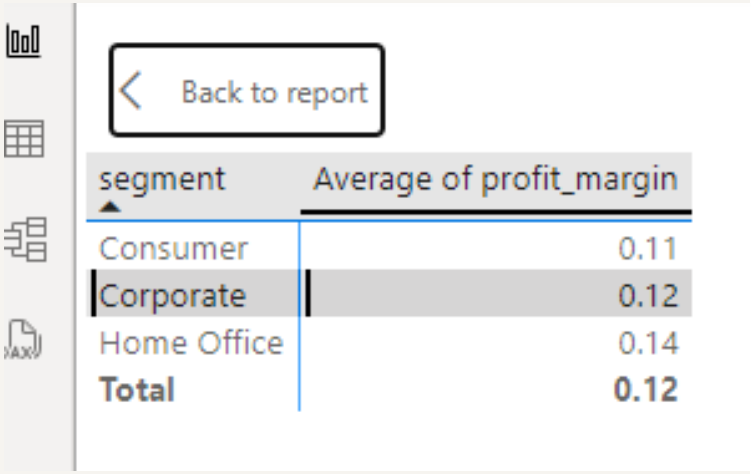
Treemap

- Select is greater than 1000
- Apply filter

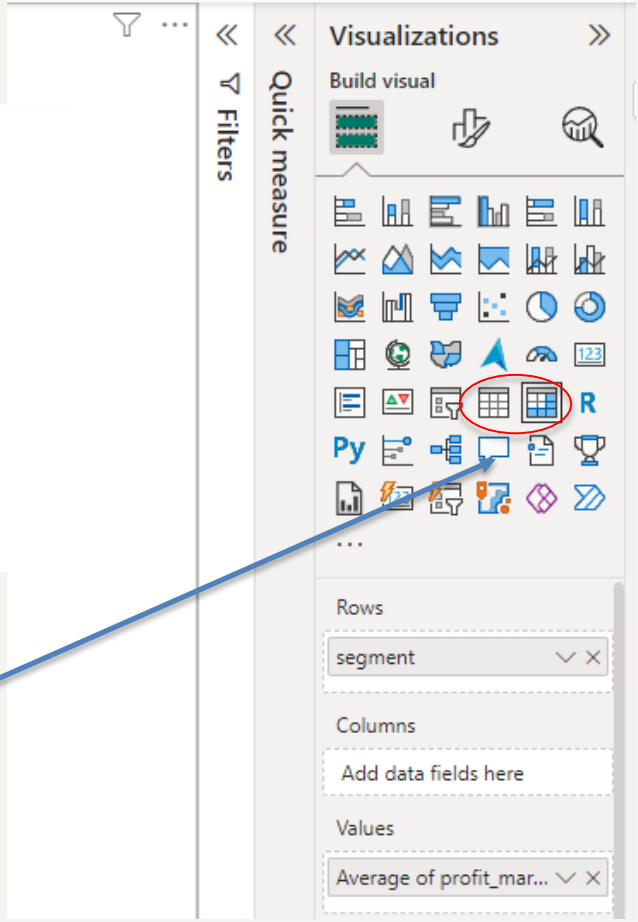


Tables and matrices in Power BI
Average profit margin per segment

- Segment is in rows
- Average profit margin in values



segment	Average of profit_margin
Consumer	0.11
Corporate	0.12
Home Office	0.14
Total	**0.12**



The icons for tables and matrices are here

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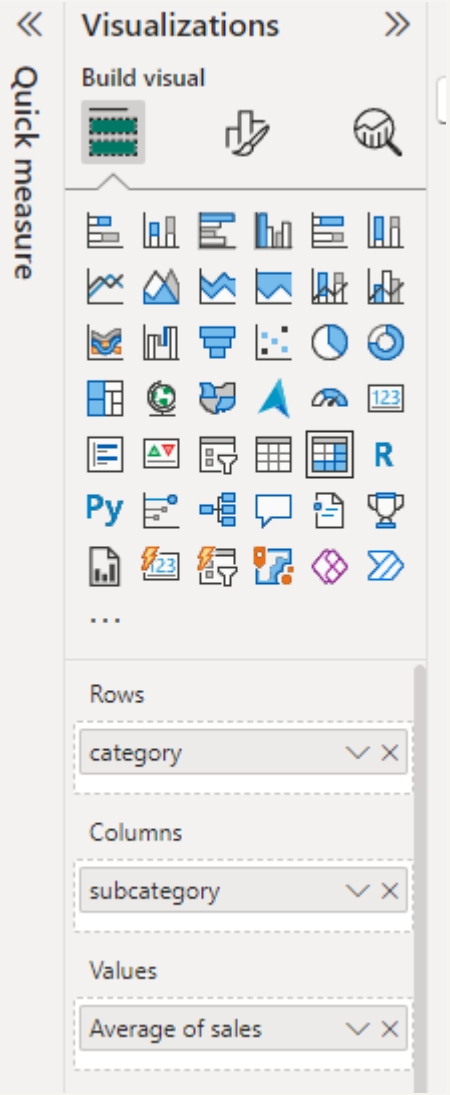
Example of

- Category is in rows
- Subcategory in columns
- Average sales in values

Back to report

category	Accessories	Appliances	Art	Binders	Bookcases	Chairs	Copiers	Envelopes	F
Furniture					503.86	532.33			
Office Supplies		230.76	34.07	133.56				64.87	
Technology	215.97						2,198.94		
Total	**215.97**	**230.76**	**34.07**	**133.56**	**503.86**	**532.33**	**2,198.94**	**64.87**	



- You can use the slider below the matrix to see all columns

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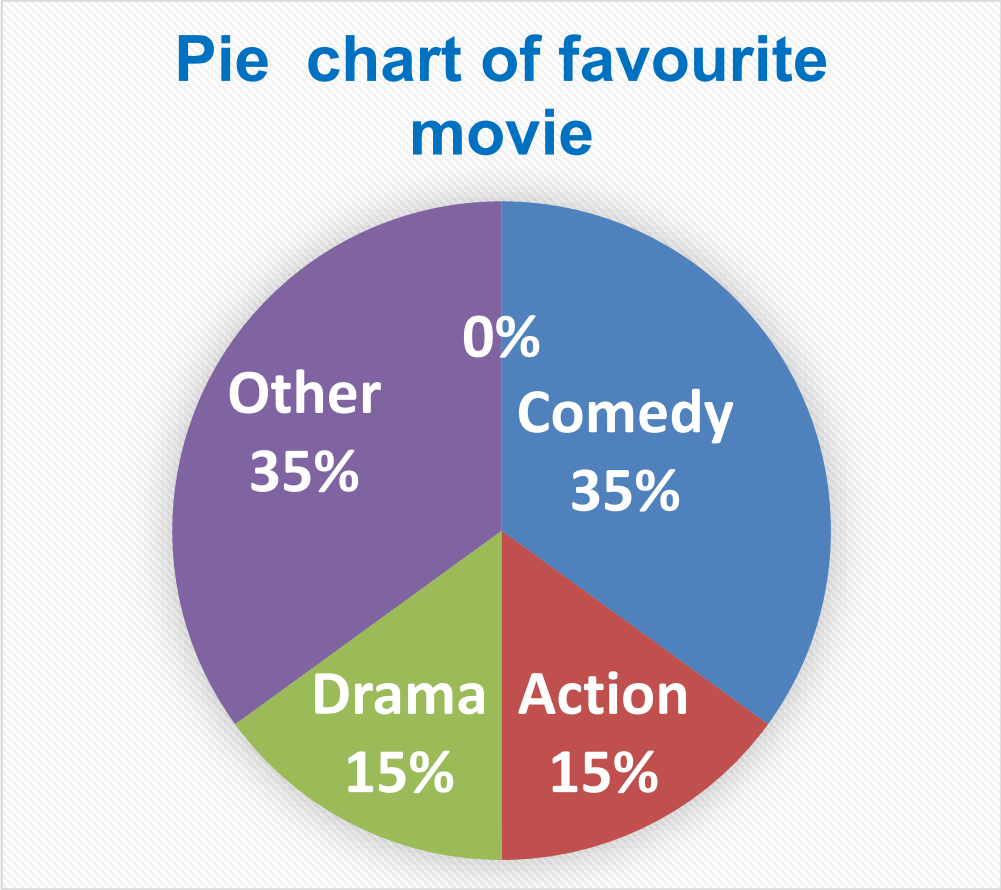
Pie chart

- Is about the "parts-to-whole" characteristic of a data set.

- Place categories in segments of a pie

- The pie must cover all categories.



- If using relative frequency %,must check total sum is 100%.

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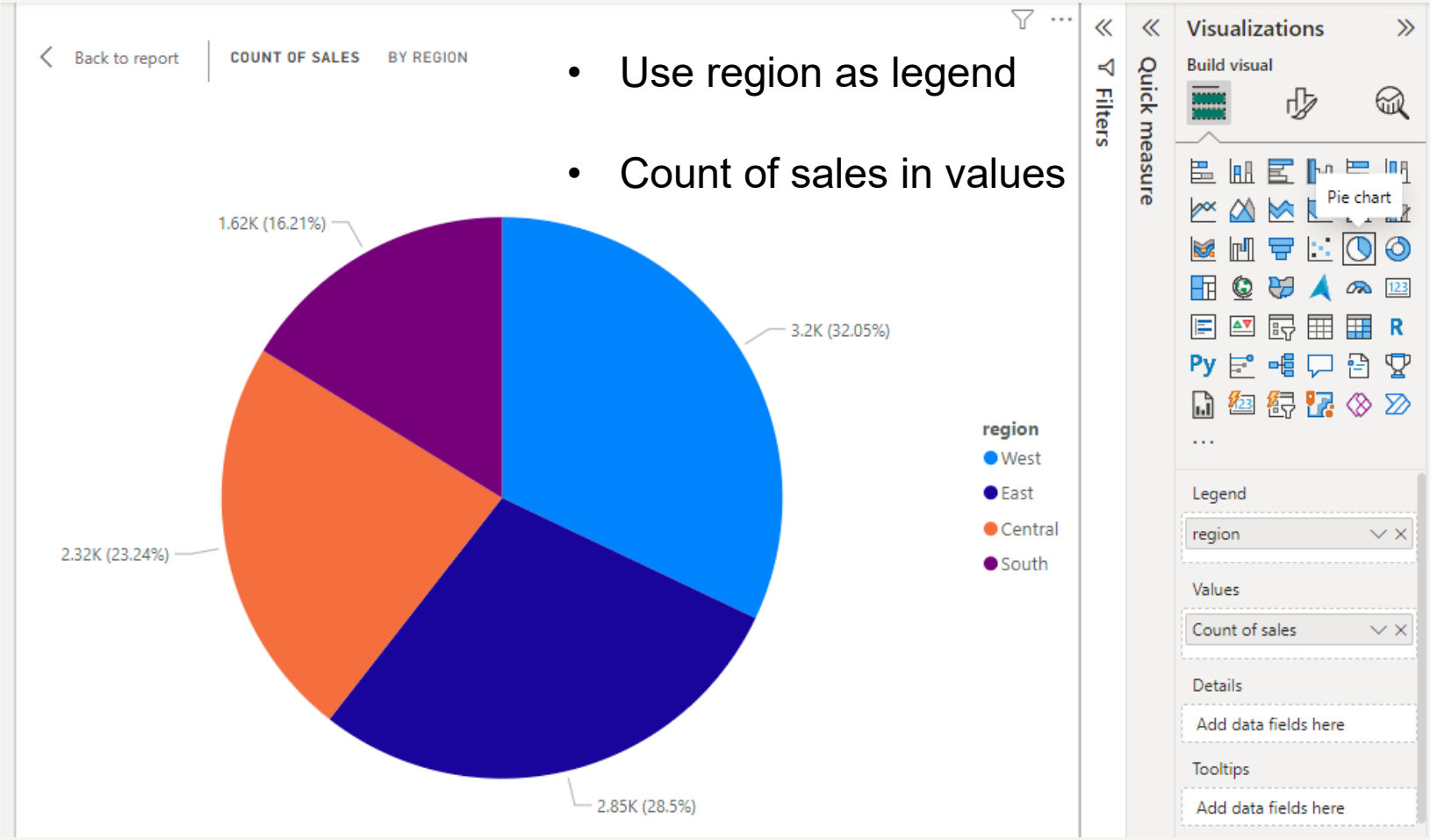
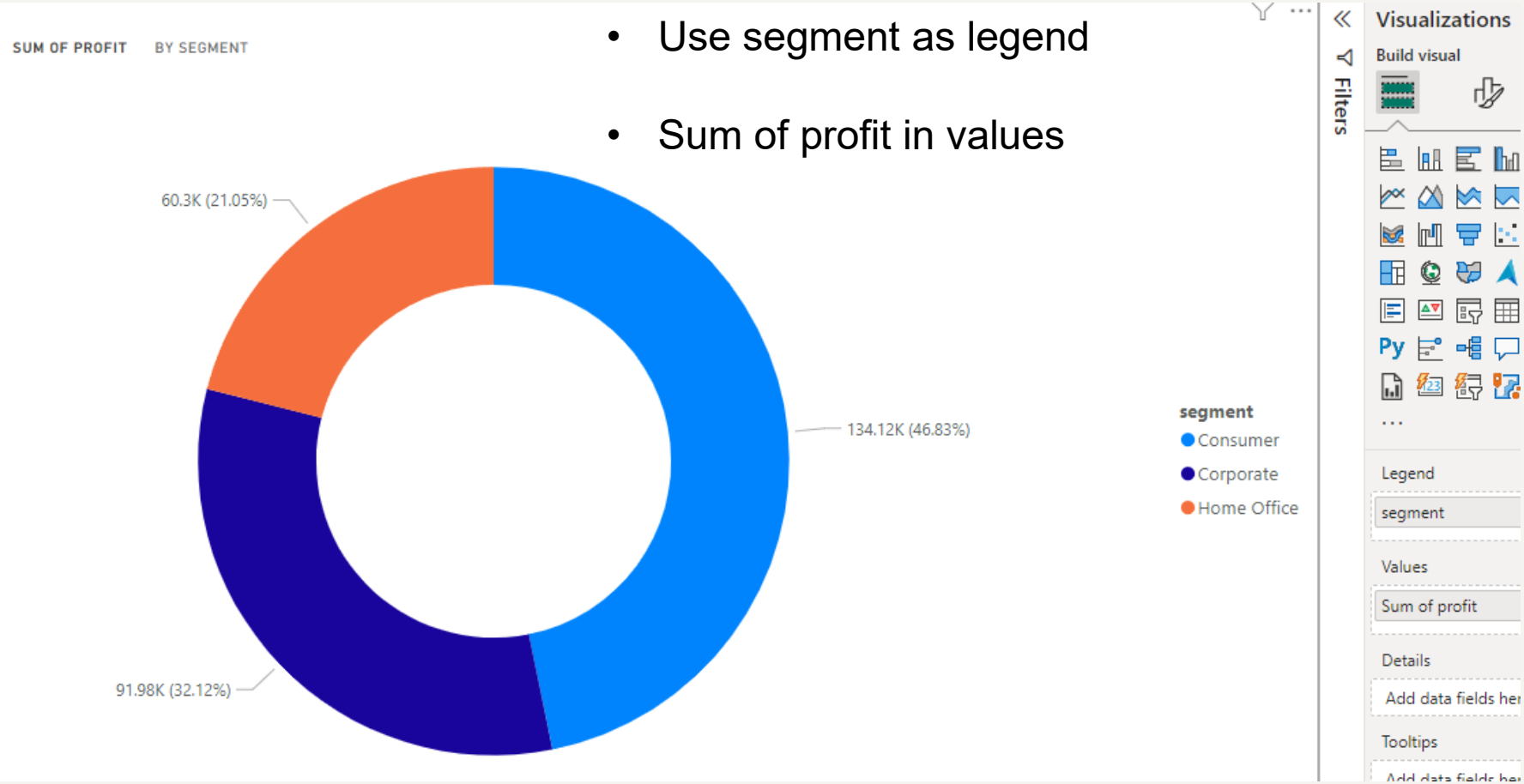
Pie Chart



- Use region as legend
- Count of sales in values

- The percent has been calculated for you. You can check by switching back to a table and calculating item count/total*100

Doughnut Chart



- Use segment as legend
- Sum of profit in values

- The percent has been calculated for you. You can check by switching back to a table and calculating item count/total*100

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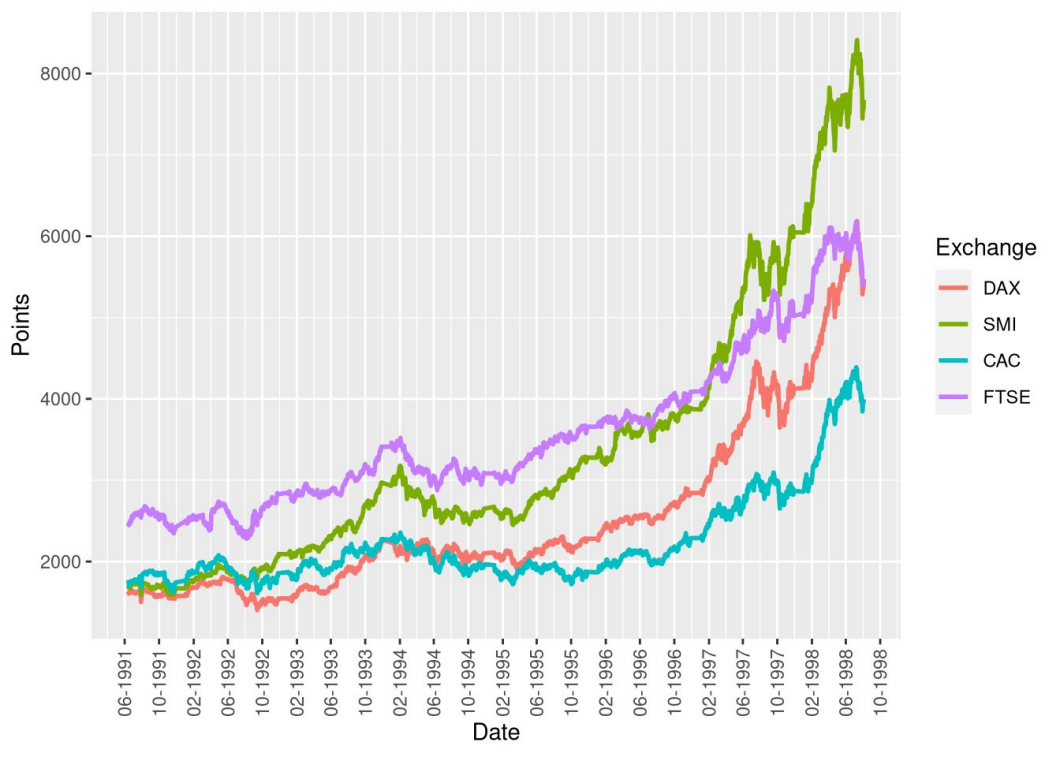
Time series

Time-Series data: is a set of time ordered observations of a quantitative characteristic of a phenomenon taken at equidistant points of time. (e.g Daily closing price of a specific stock for the last 10 years)



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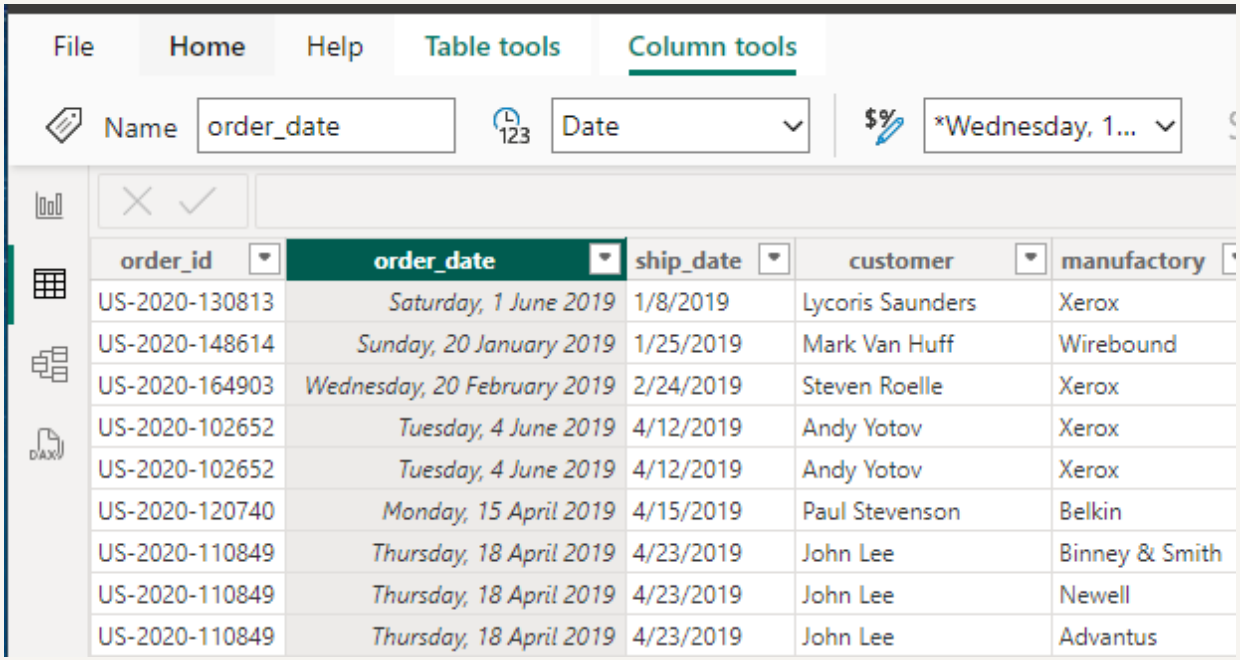
Time series

- Time-Series of sales by date
- Currently Order date is text so click on the table icon on the LHS panel
- Select the order date column and change text to date



order_id	order_date	ship_date	customer	manufactory
US-2020-130813	Saturday, 1 June 2019	1/8/2019	Lycoris Saunders	Xerox
US-2020-148614	Sunday, 20 January 2019	1/25/2019	Mark Van Huff	Wirebound
US-2020-164903	Wednesday, 20 February 2019	2/24/2019	Steven Roelle	Xerox
US-2020-102652	Tuesday, 4 June 2019	4/12/2019	Andy Yotov	Xerox
US-2020-102652	Tuesday, 4 June 2019	4/12/2019	Andy Yotov	Xerox
US-2020-120740	Monday, 15 April 2019	4/15/2019	Paul Stevenson	Belkin
US-2020-110849	Thursday, 18 April 2019	4/23/2019	John Lee	Binney & Smith
US-2020-110849	Thursday, 18 April 2019	4/23/2019	John Lee	Newell
US-2020-110849	Thursday, 18 April 2019	4/23/2019	John Lee	Advantus

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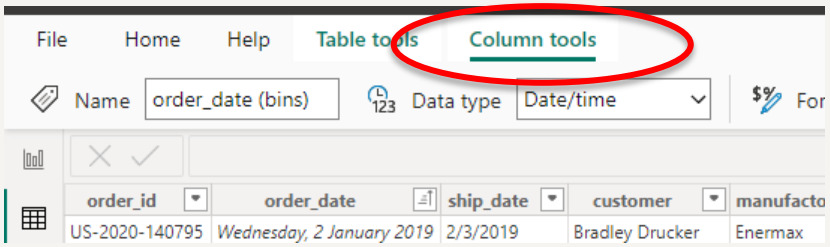
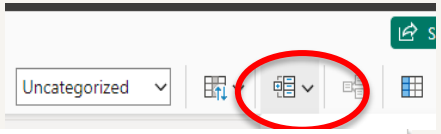
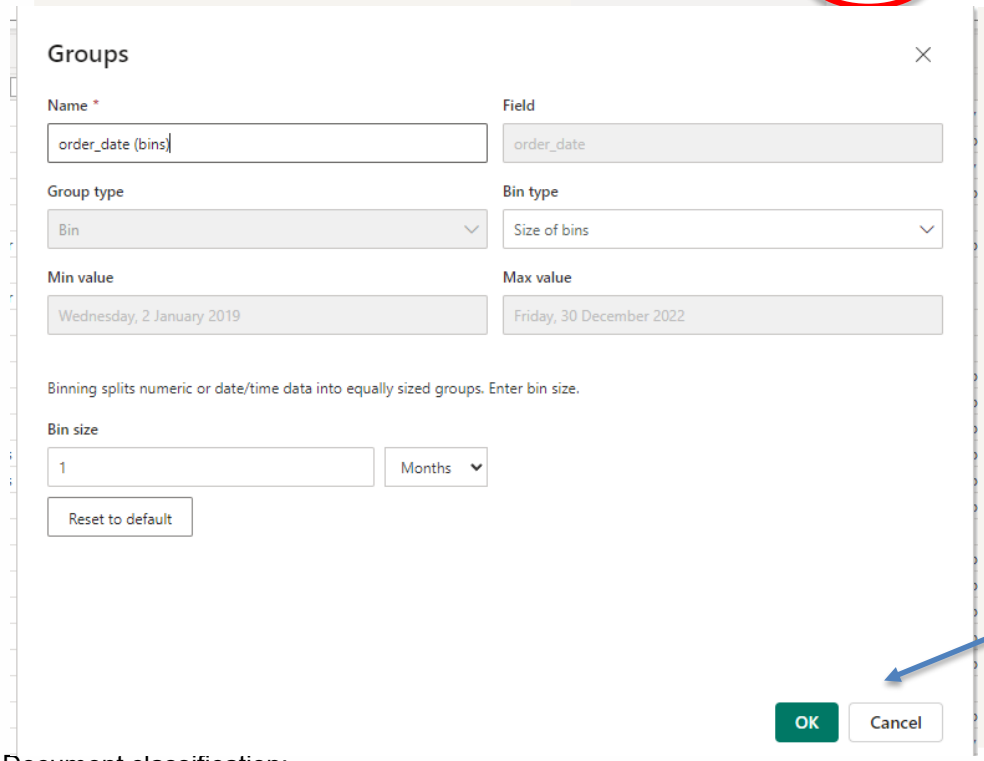
Create a new column where dates are grouped as Jan, Feb……







- Select column tools
- Go to the groups icon
- The following will create the new column for you
- Press OK

Old and new date column, plot

	order_date
5	Wednesday, 2 January 2019
9	Thursday, 3 January 2019
7	Thursday, 3 January 2019
0	Thursday, 3 January 2019
9	Thursday, 3 January 2019
2	Thursday, 3 January 2019
0	Thursday, 3 January 2019
2	Thursday, 3 January 2019
9	Thursday, 3 January 2019
9	Thursday, 3 January 2019
9	Thursday, 3 January 2019
1	Friday, 4 January 2019
1	Friday, 4 January 2019
9	Friday, 4 January 2019
9	Friday, 4 January 2019
0	Sunday, 6 January 2019

order_date (bins)
January 2019
January 2019
January 2019
January 2019
January 2019
January 2019
January 2019
January 2019
January 2019
January 2019
January 2019
January 2019
January 2019
January 2019
January 2019

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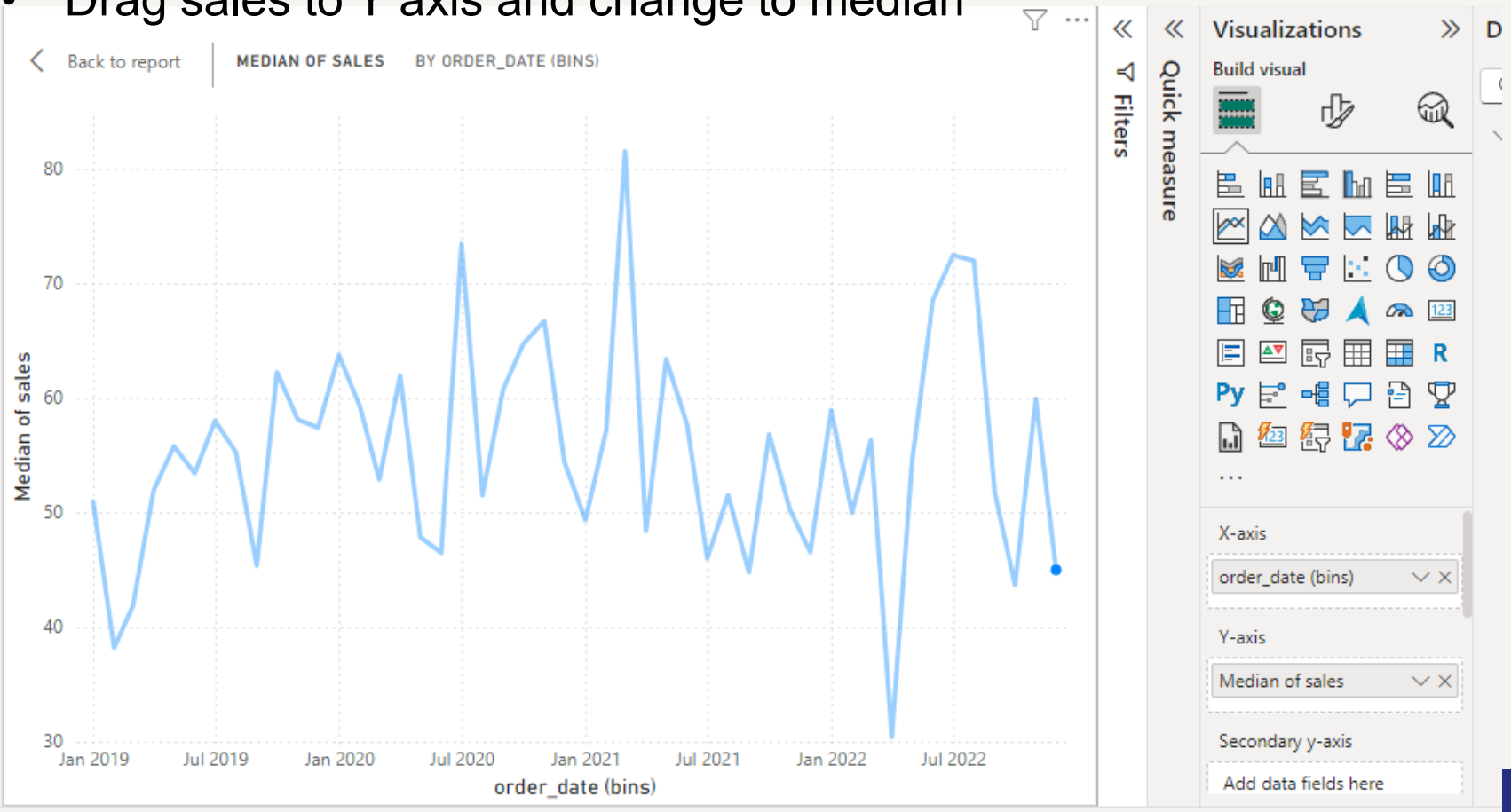
Time series of median sales

- Select line chart
- Drag order date (bins) to X axis
- Drag sales to Y axis and change to median



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Correlation

Correlation measures the *strength* of the *linear* relationship between *two* variables.

Drawbacks:

- Strength
  - not the slope
- Linear
  - misses nonlinearities completely
- Limited
  - Relationship between only two variables
    - Shows only “shadows” of multidimensional relationships

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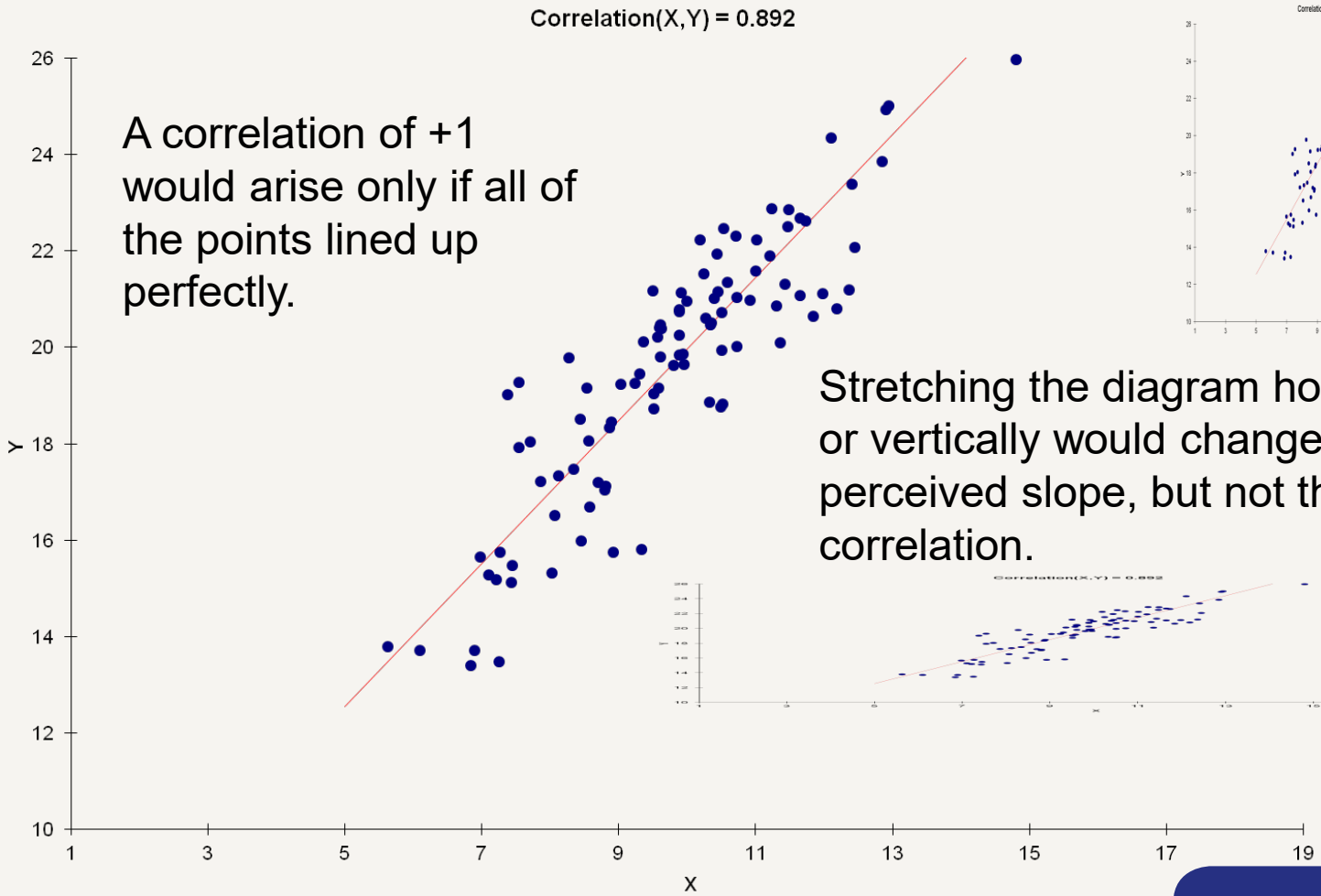
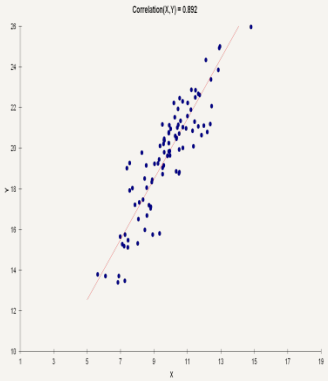
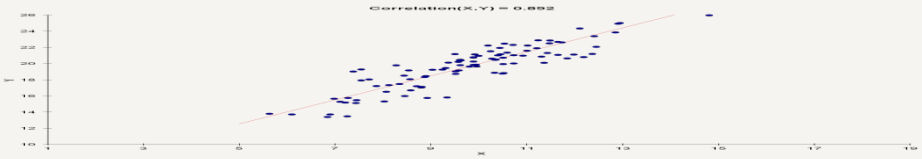
Example: Correlation



A correlation of +1 would arise only if all of the points lined up perfectly.

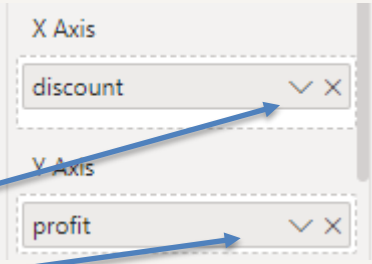


Stretching the diagram horizontally or vertically would change the perceived slope, but not the correlation.

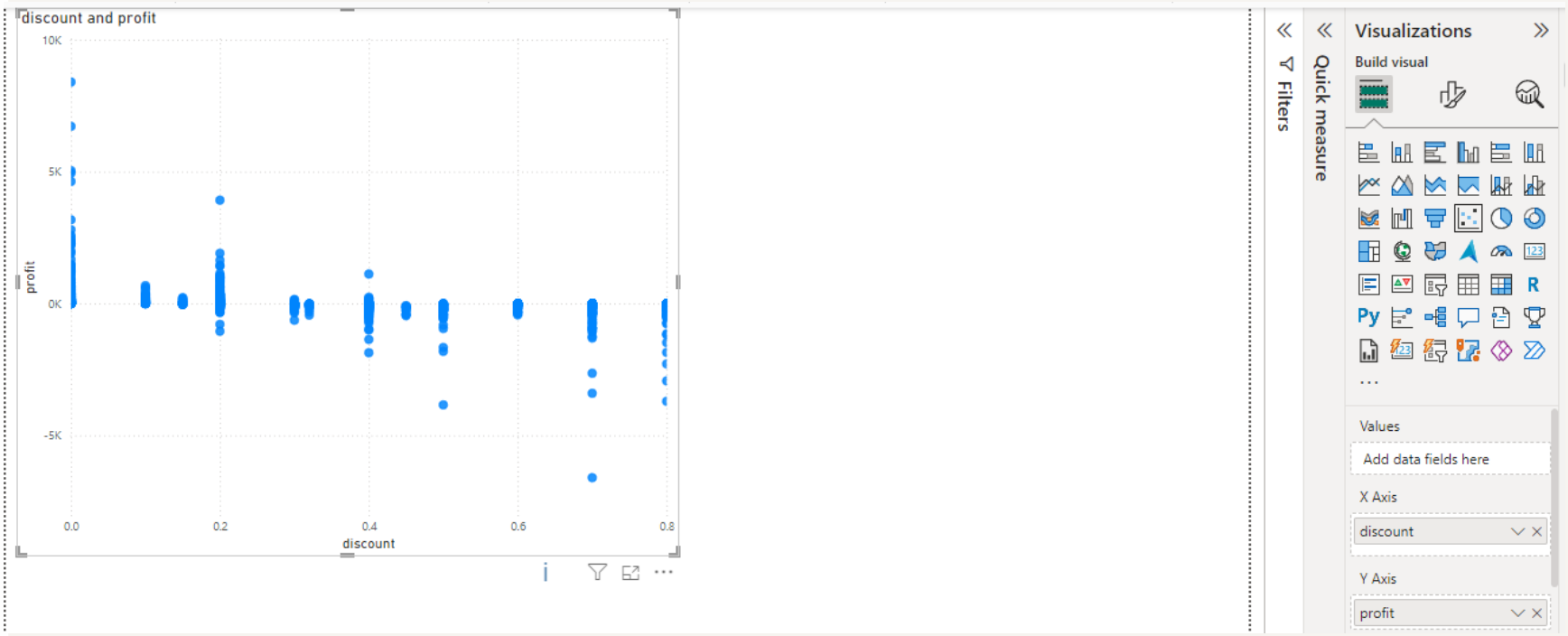


Correlation

- Select scatter
- Drag sum of discount to X axis
- Drag sum of profit

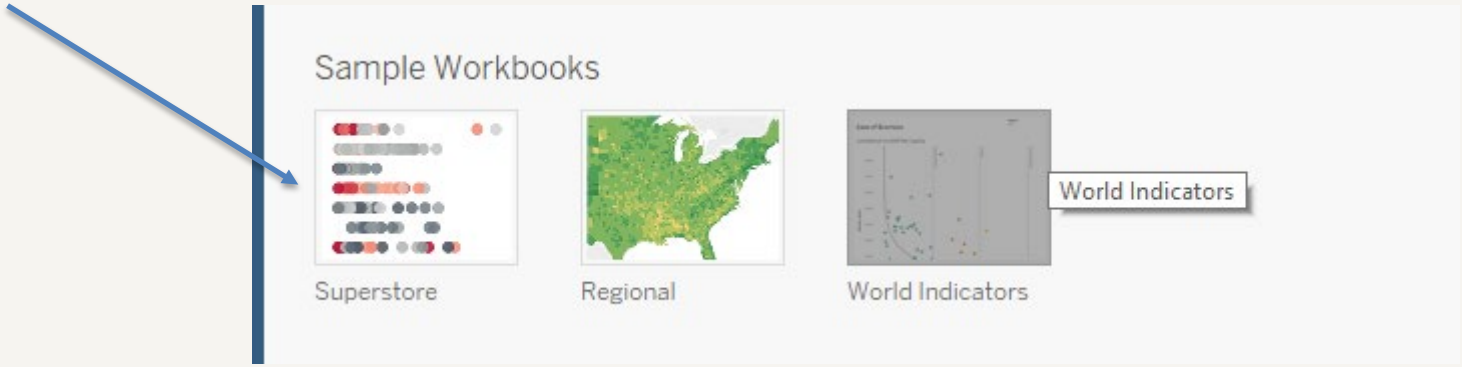


- Change both to “Don’t summarize” by clicking on the arow next to the var name



Charts in Tableau

- The superstore data is one of the sample data set in Tableau
- Open Tableau and it should appear in the sample workbooks,



- if not use the connect to a file (LHS pane) to connect to the CSV

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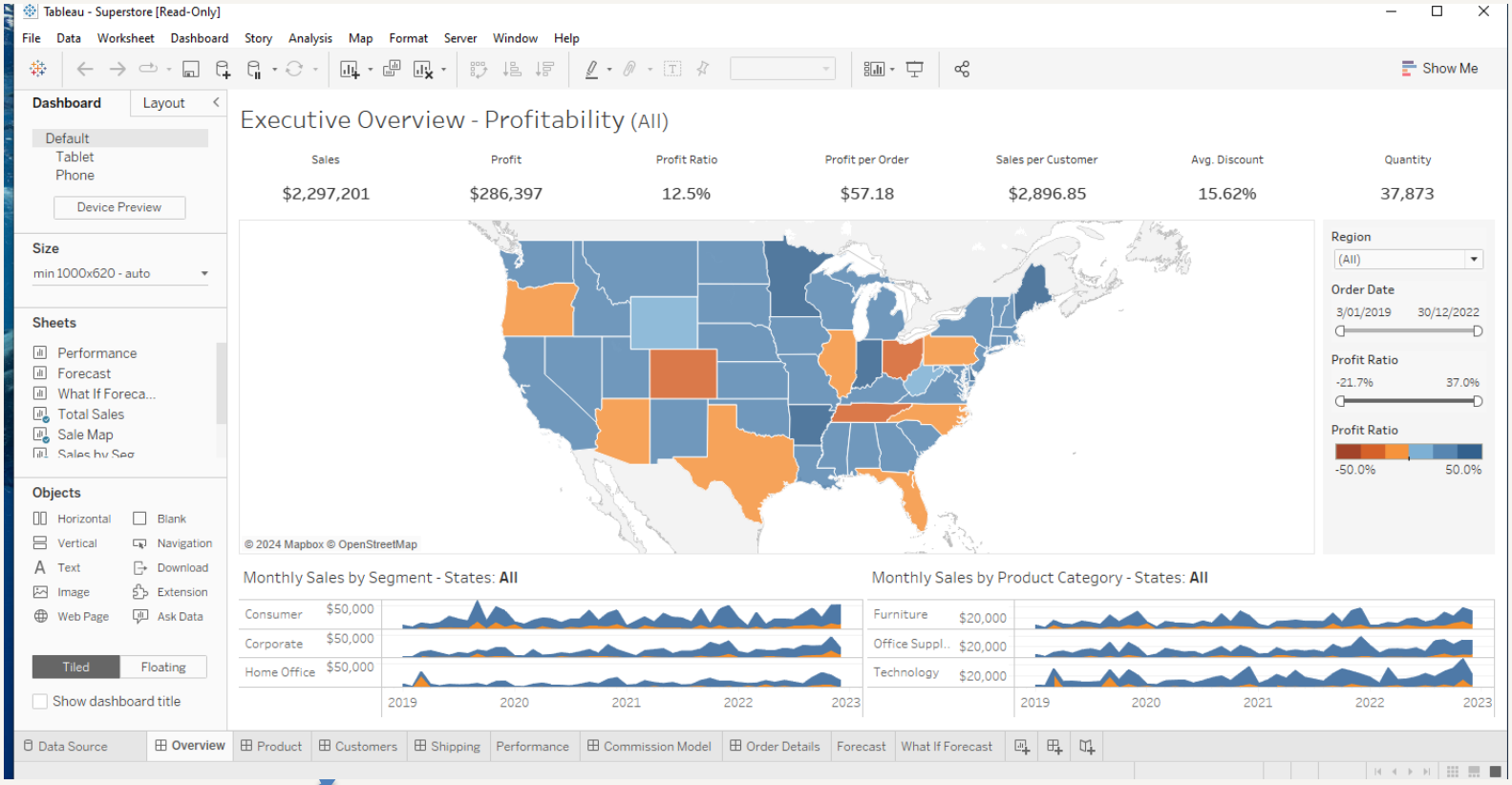
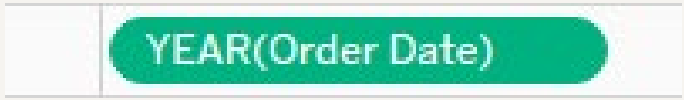
Charts in Tableau



- If you can access the Superstore sample workbook, examples of charts can be accessed by clicking on the pages at the bottom of the Tableau window. You can open a new sheet and try your own.

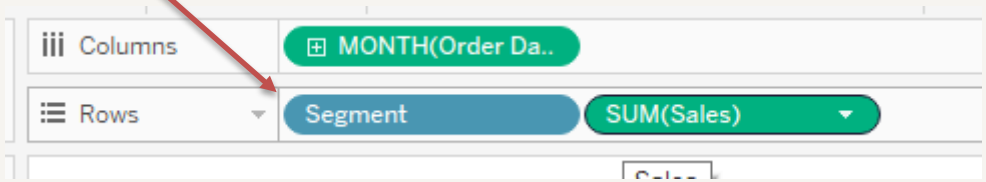
Simple Time series and Forecasting in Tableau

- Create the initial time series using the sample data in Tableau
- Drag Year(order date to columns)



- Change it to continuous and select year in the second menu as you scroll down
- Change to month by clicking on the plus sign

- Drag Segment and sum of sales to rows



- Drag Segment to the colour box (under Marks)

See chart next page

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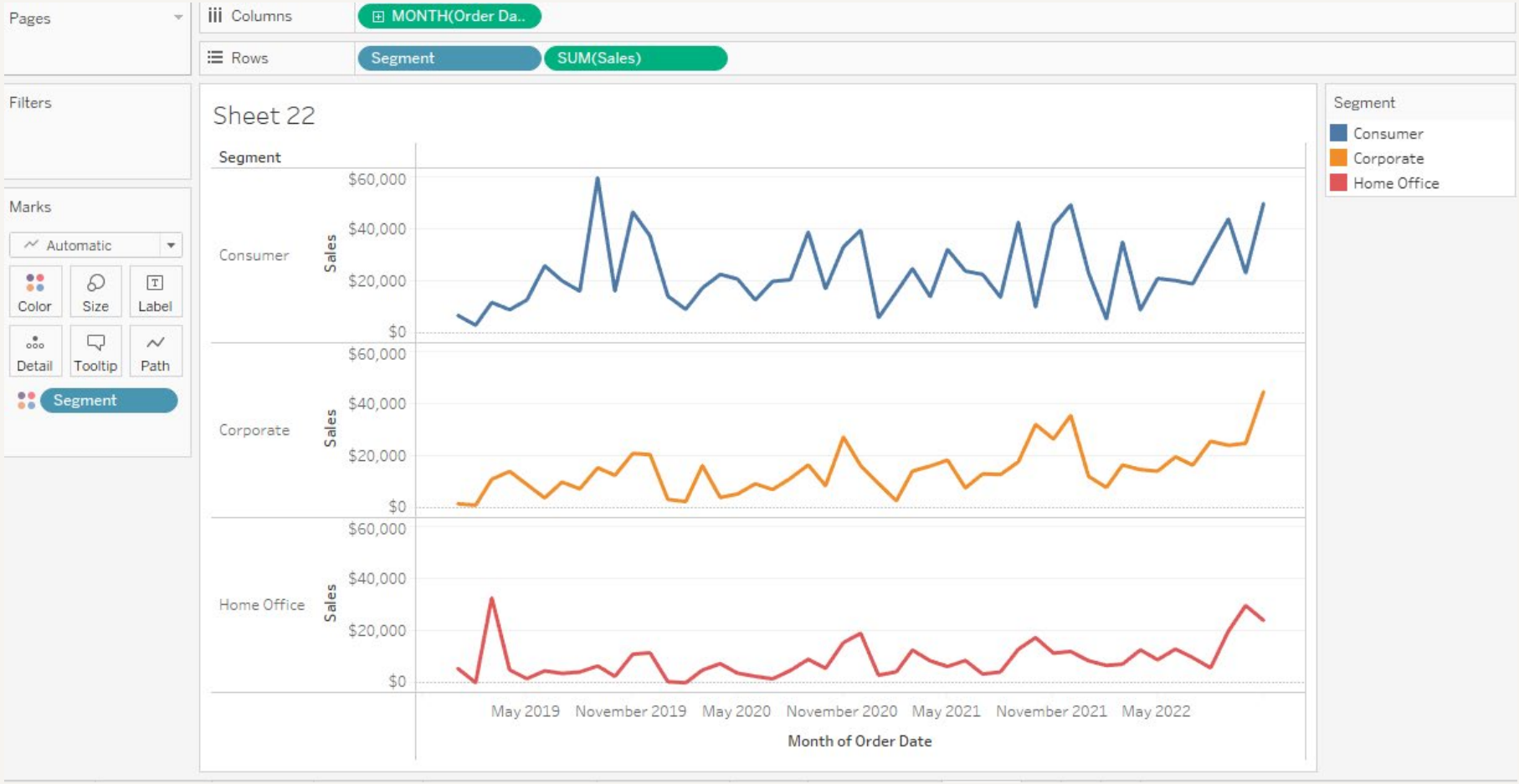
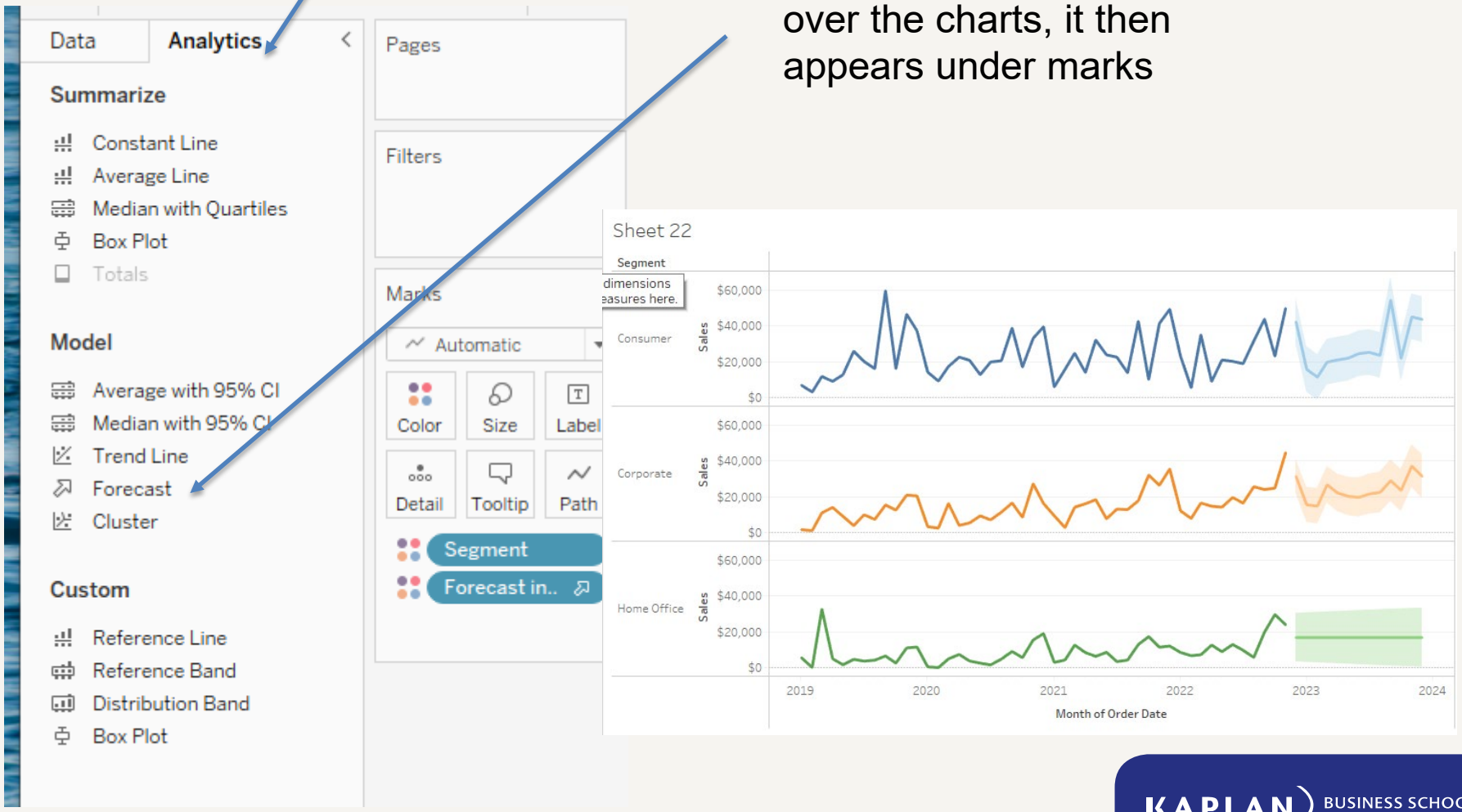
Time series in Tableau



Simple Forecast

Go to the Analytics pane (LHS)

Drag the forecast indicator over the charts, it then appears under marks



Activity: Interpret the results

- Review your initial questions and those supplied for you.
- If the current charts didn't answer the questions, try creating more charts in either Power BI and/or Tableau and find the answers.
- Experiment with the colours, etc.
- Try working as a small group
- Report back to class

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Charting will continue

- We create more charts and tables in future weeks
- Histograms in Power BI require you to click on the three dots in the visuals pane and get more visuals
- Try charting at home

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